# EART 22201 Igneous minerals and processes

## Igneous processes: Crystallization

Lesson 4




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# Igneous processes: Crystallization

**1.1 From solid to liquid**

1.2 Equilibrium and fractional crystallization

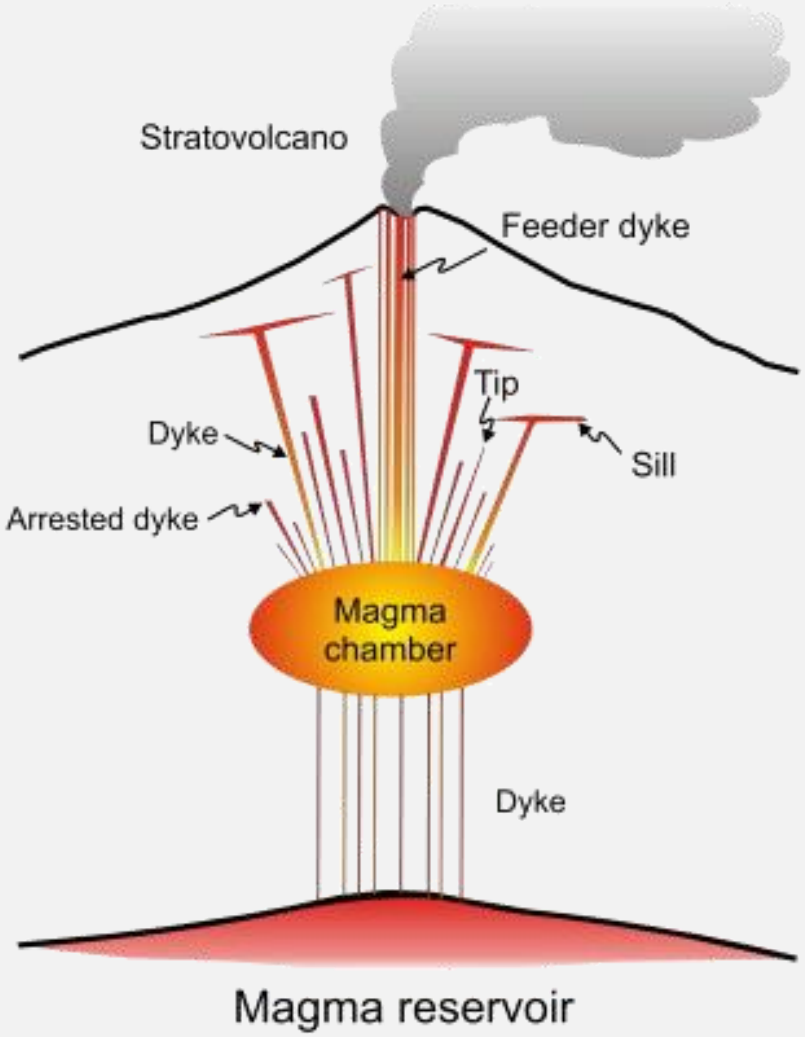
1.3 Bowen's reaction series





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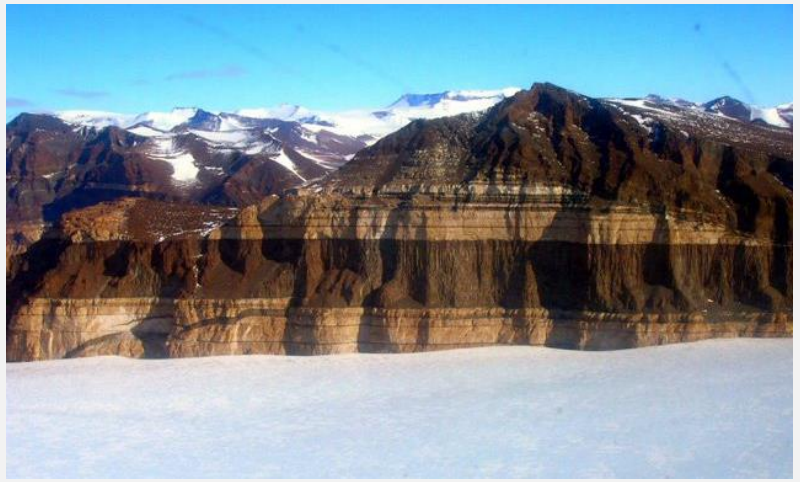
# From solid to liquid: where does crystallization happen?




Extrusive environments e.g. lavas at Earth's surface

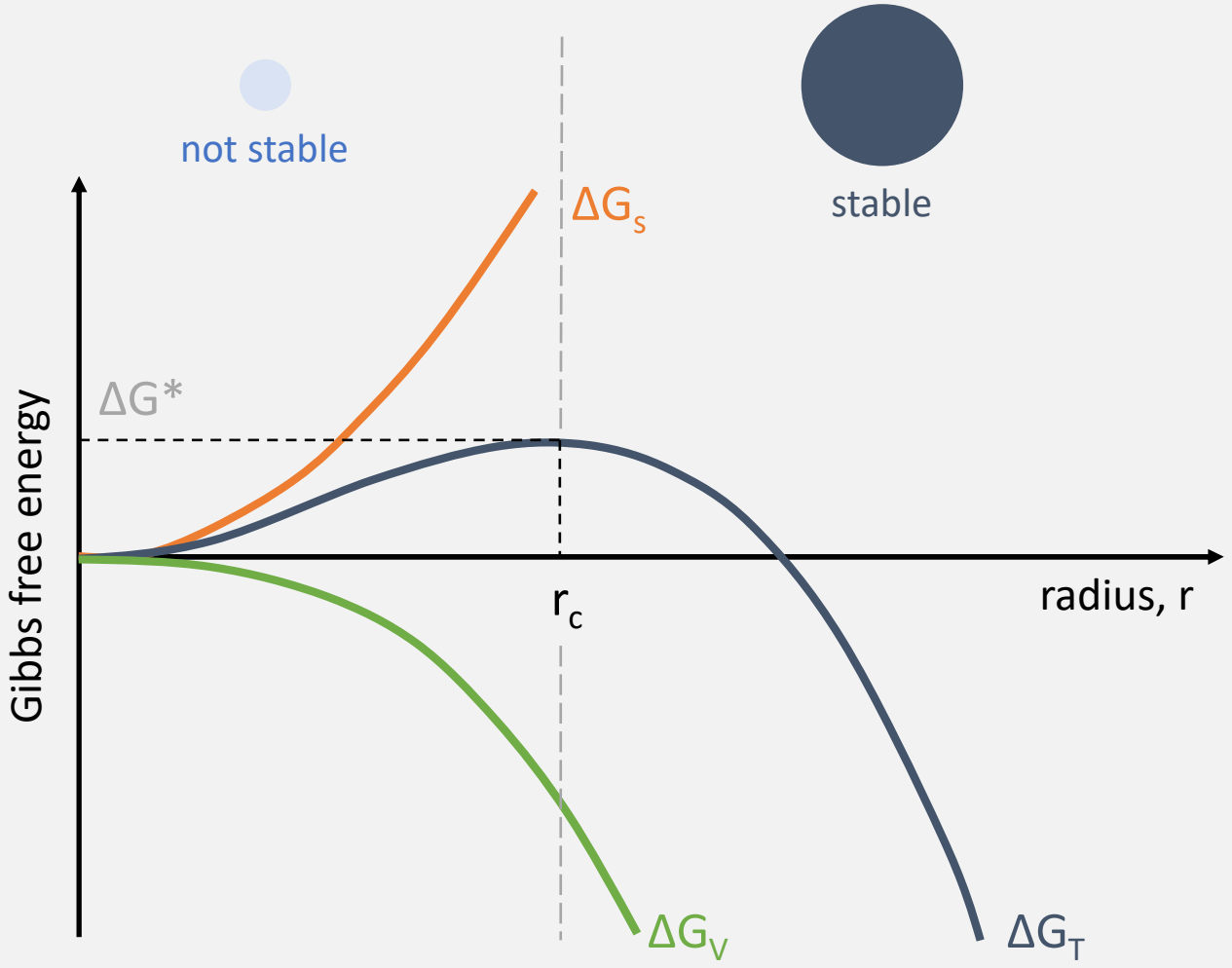


Intrusive environments e.g. sills within the crust




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# Nucleation: the formation of a stable new crystal and the start of crystallization




The total Gibbs free energy $\Delta G_T$ of a crystal is the sum of its surface energy $\Delta G_s$ and its volume energy $\Delta G_V$.

The total free energy increases until the critical radius $r_c$ is reached, after which further growth causes $\Delta G_T$ to decrease.

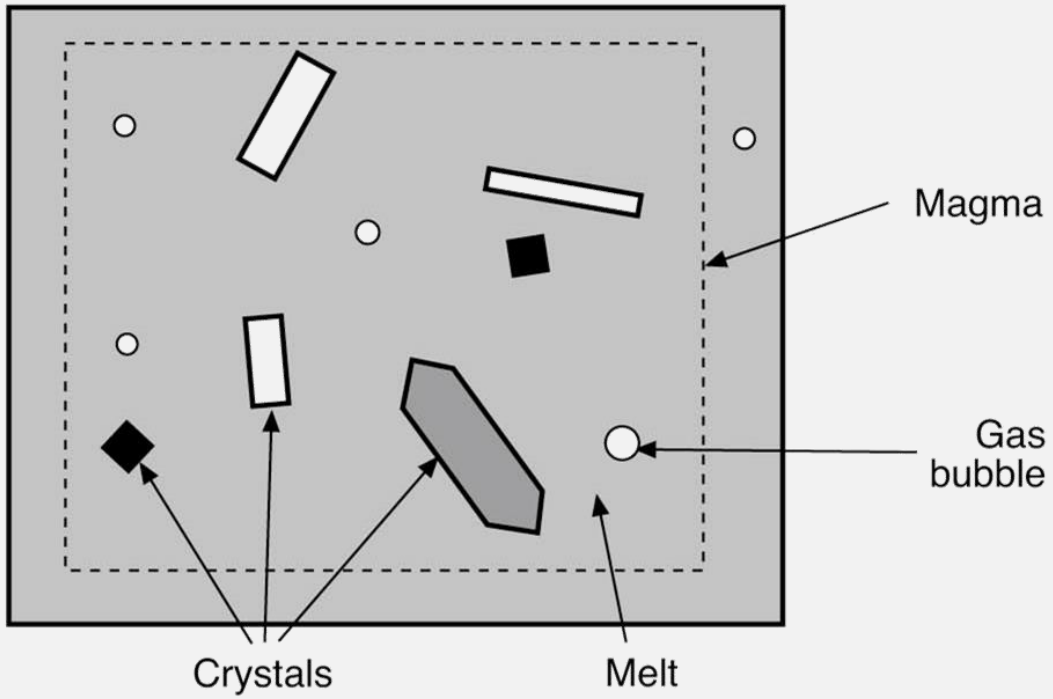
A crystal is stable once it has reached the critical radius. $\Delta G^*$ is the activation energy required to form a stable crystal nucleus.

Crystals smaller than the critical radius may continue to grow, or may be resorbed back into the melt.

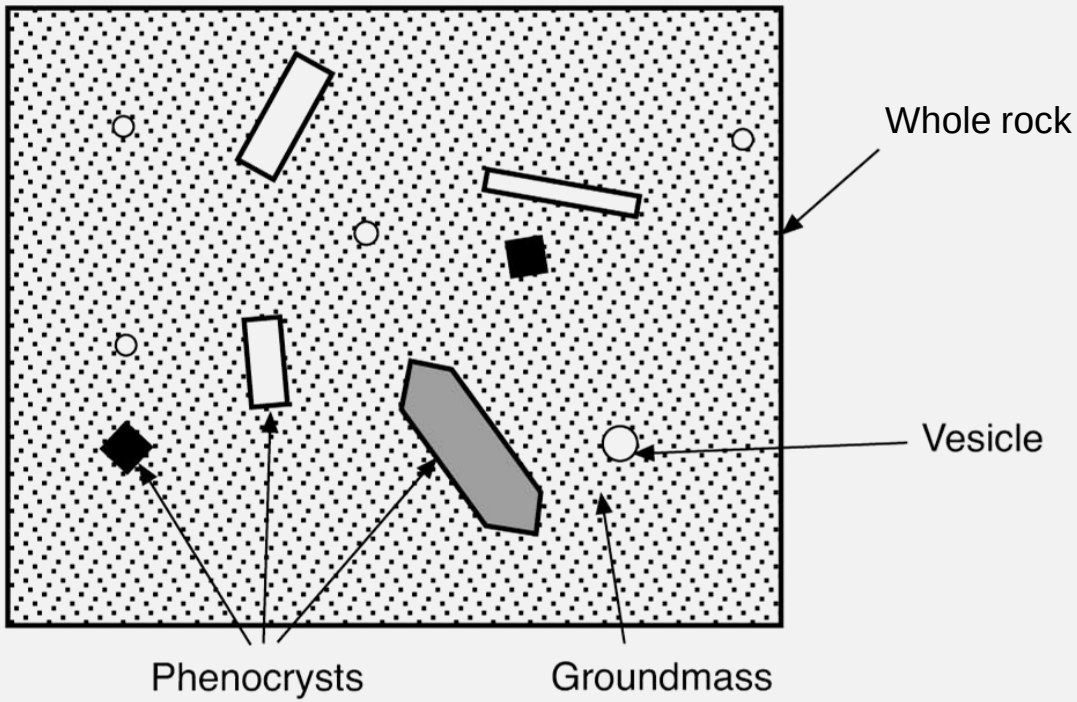

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# Some terminology for describing crystallizing magmas and the resulting rocks

Molten (magma)




Crystalline (solid)





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# Igneous processes: Crystallization

1.1 From solid to liquid

**1.2 Equilibrium and fractional crystallization**

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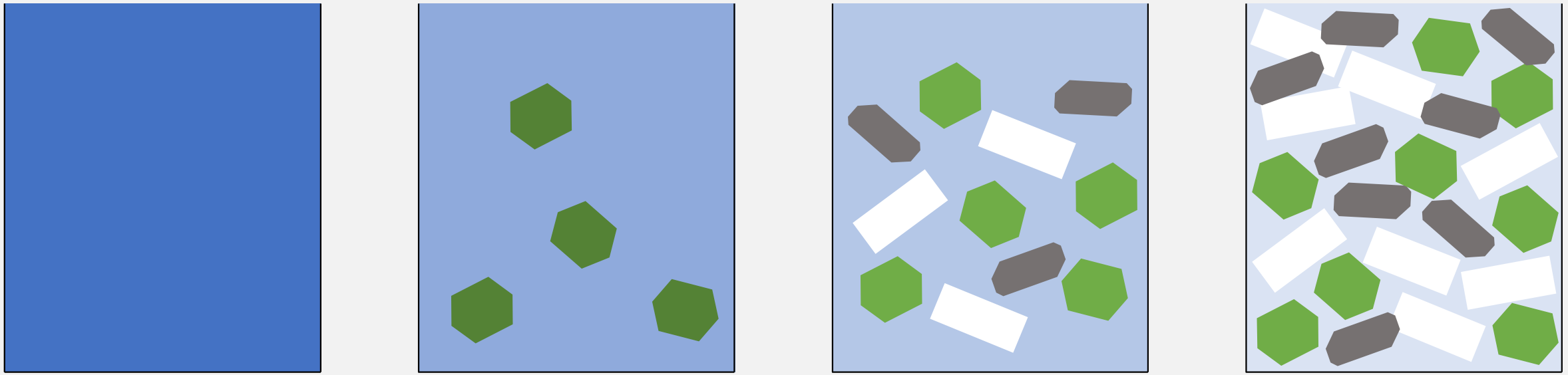




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# Equilibrium crystallization

See also EART 26201 Principles of Geochemistry, Lecture 1



100% melt
0% crystallized

90% melt
10% crystallized

50% melt
50% crystallized

20% melt
80% crystallized

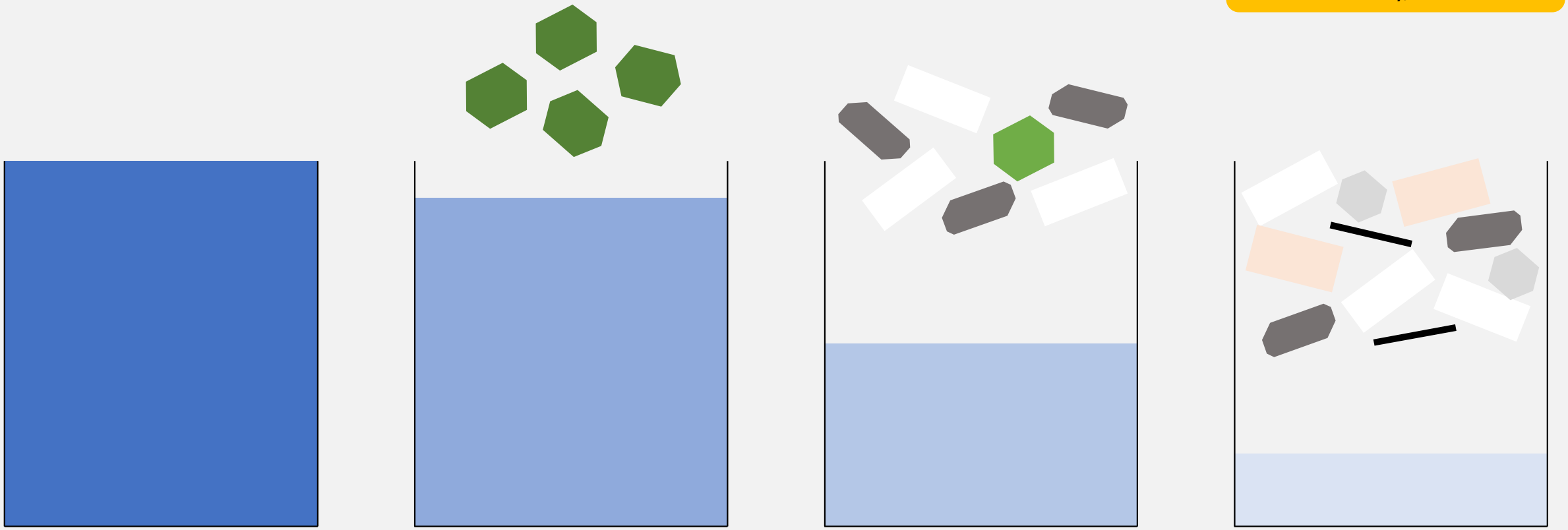
cooling



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# Fractional crystallization

See also EART 26201 Principles of Geochemistry, Lecture 1



100% melt
0% crystallized

100% melt
10% crystallized

100% melt
50% crystallized

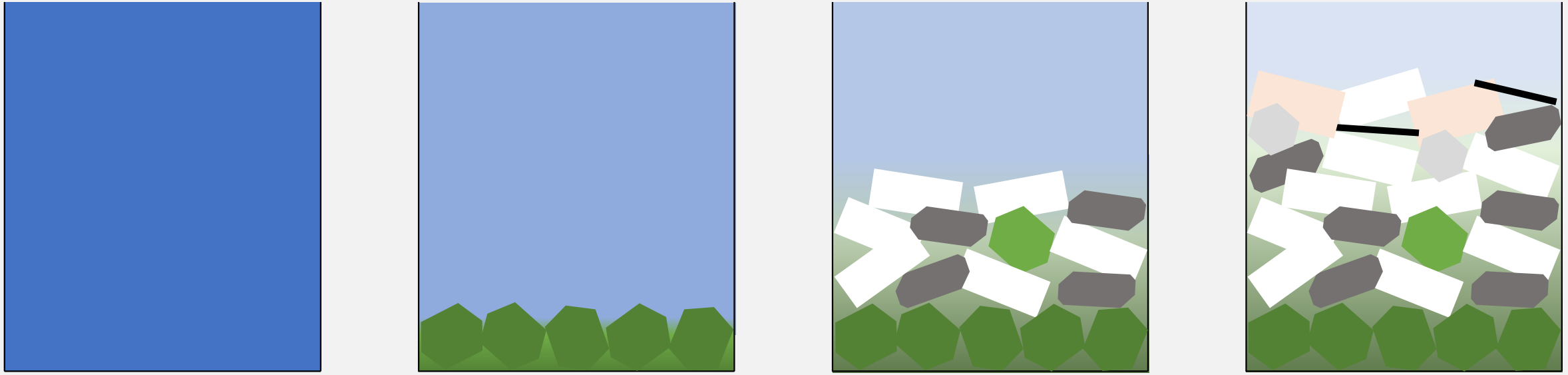
100% melt
80% crystallized

cooling



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# Closed system crystallization



100% melt
0% crystallized

90% melt
10% crystallized

50% melt
50% crystallized

20% melt
80% crystallized

cooling



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# Igneous processes: Crystallization

1.1 From solid to liquid

1.2 Equilibrium and fractional crystallization

**1.3 Bowen's reaction series**

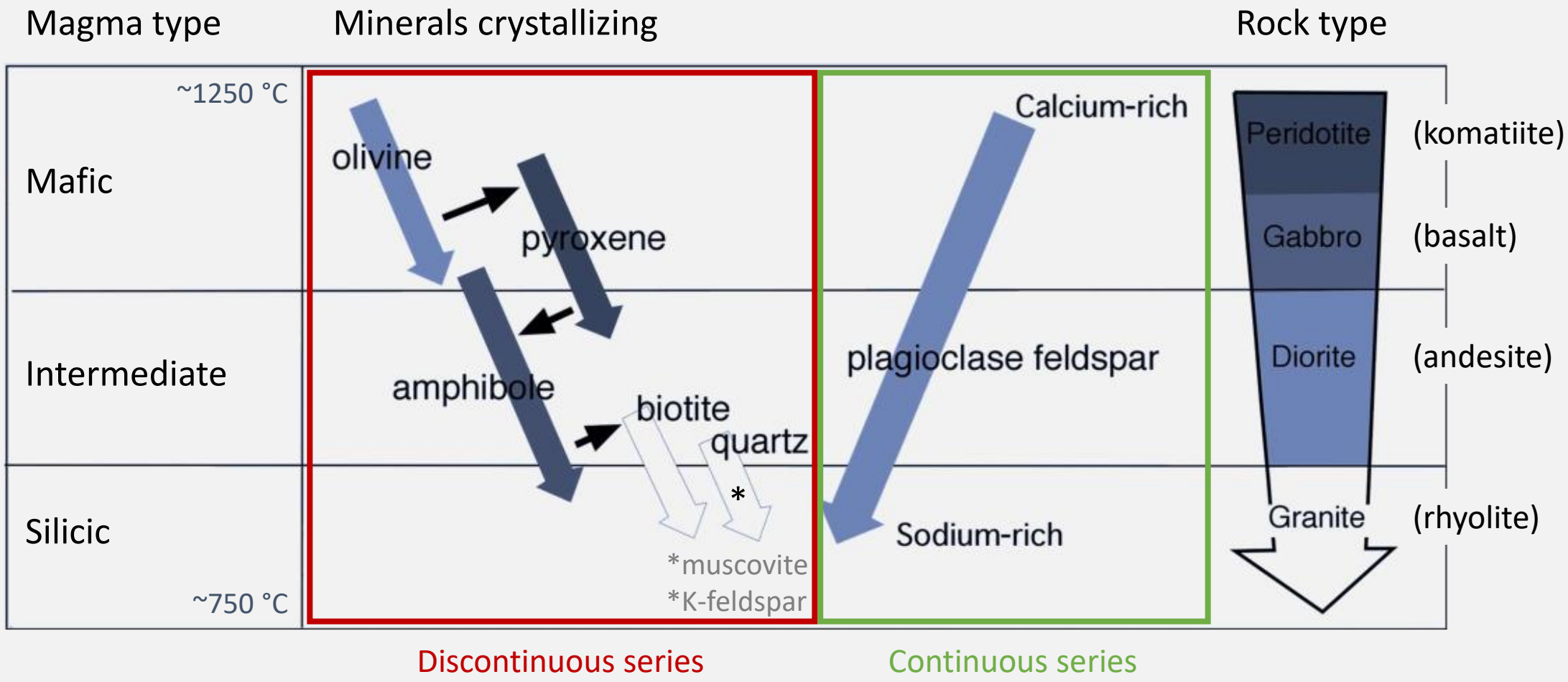




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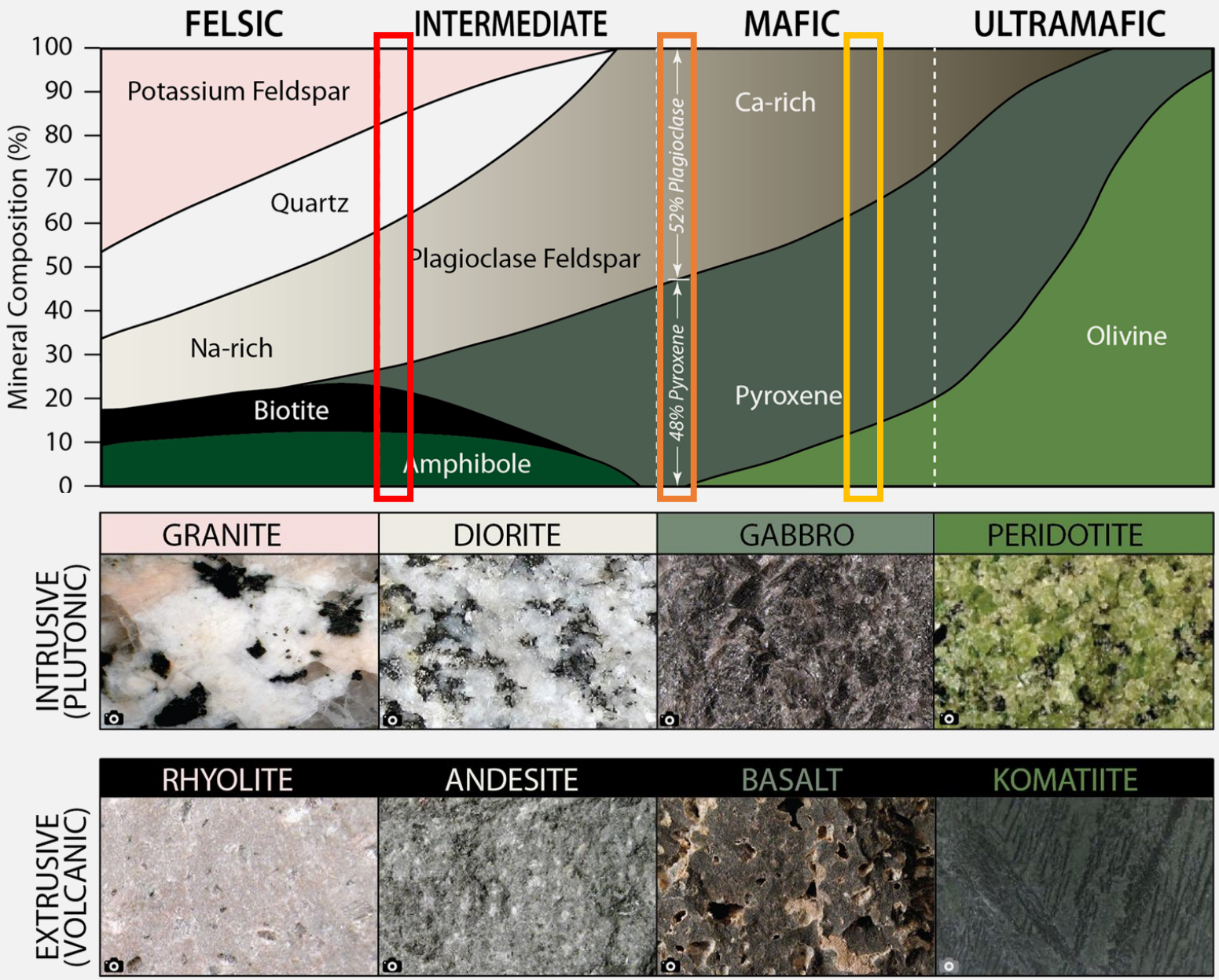
# Bowen's reaction series

See also EART 26201 Principles of Geochemistry, Lecture 1





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Crystallization of mafic magma 1
~10% olivine
~50% pyroxene (augite)
~40% plagioclase (calcic)

Crystallization of mafic magma 2
~50% pyroxene (augite)
~50% plagioclase (calcic)

Crystallization of intermediate magma 3
~10% amphibole (hornblende)
~10% biotite
~5% pyroxene
~30% plagioclase (sodic)
~25% quartz
~20% K-feldspar

A note of caution: Bowen's Reaction Series doesn't necessarily predict the order of crystallization in a magma. Need to look at the thin section!


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# Igneous processes: Crystallization

You should now be able to:

- Apply correct terminology to describe the solid, liquid and gas components of a magma
- Explain the difference between equilibrium and fractional crystallization
- Describe the key features of Bowen's reaction series
- Qualitatively describe how modal mineralogy changes as a function of magma evolution




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# Igneous processes: Crystallization

**2.1 Describing igneous hand specimens**

2.2 Mineralogical classification of igneous rocks





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# Rock description and/or rock composition enables classification

## Igneous rock description

### Mineralogical

- Grain size
- Colour
- Modal mineralogy

+

### Textural

- Shape and distribution of minerals
- Mineral alignment
- Order of crystallization

## Igneous rock classification

### Petrological

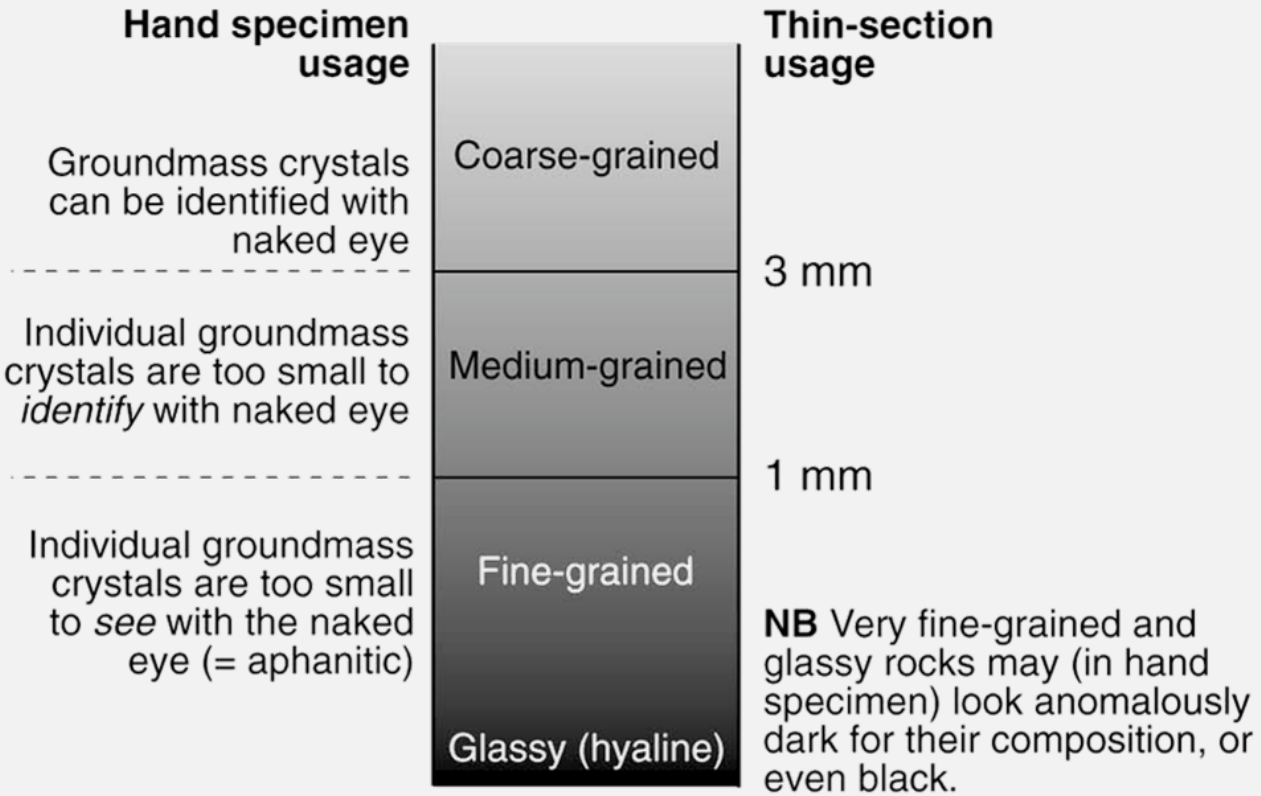
- Based on mineralogical and textural observations

### Chemical

- Based on chemical analysis of the rock


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# Describe the grain size. Give measurements wherever possible



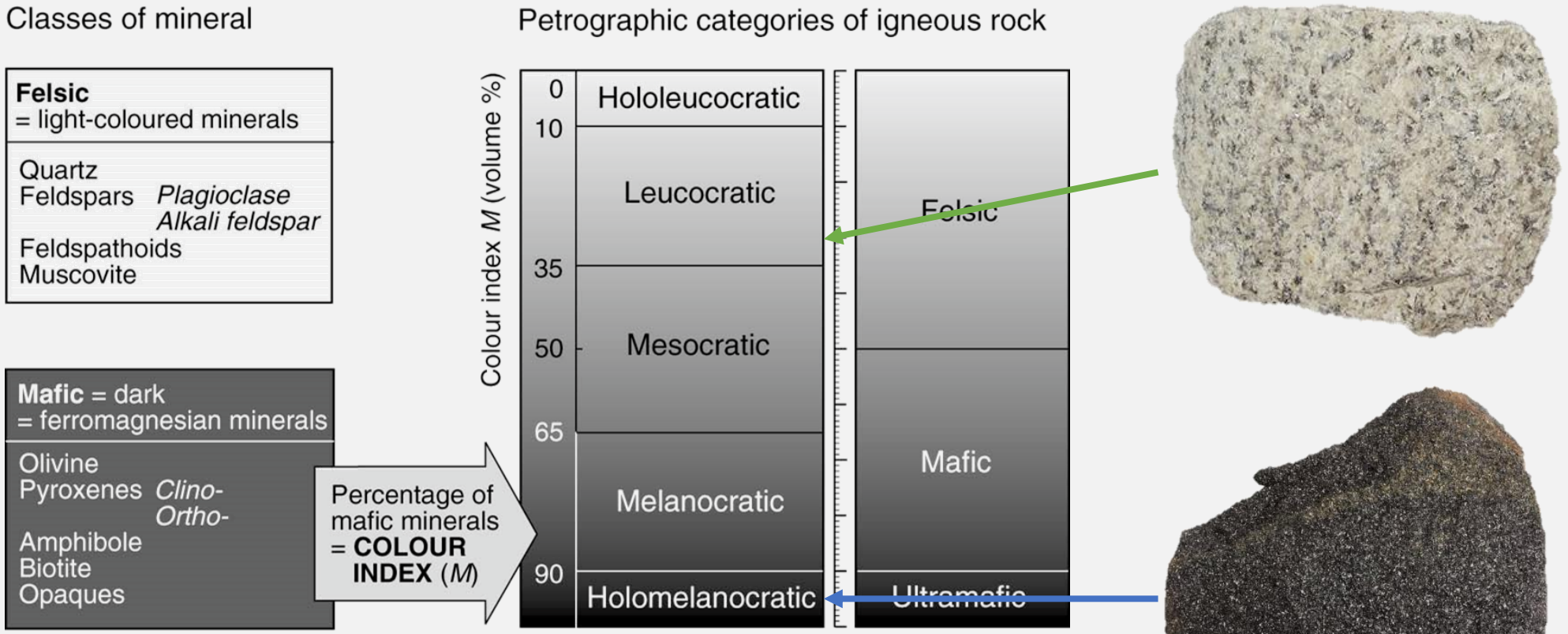

Gill (2010) Fig. 1.3



In a rock with a range of grain sizes, the groundmass grain size dictates whether the rock is named using fine-grained or coarse-grained terminology.


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# Describe the colour index using the proportions of light (felsic) to dark (mafic) minerals






leucocratic
muscovite granite

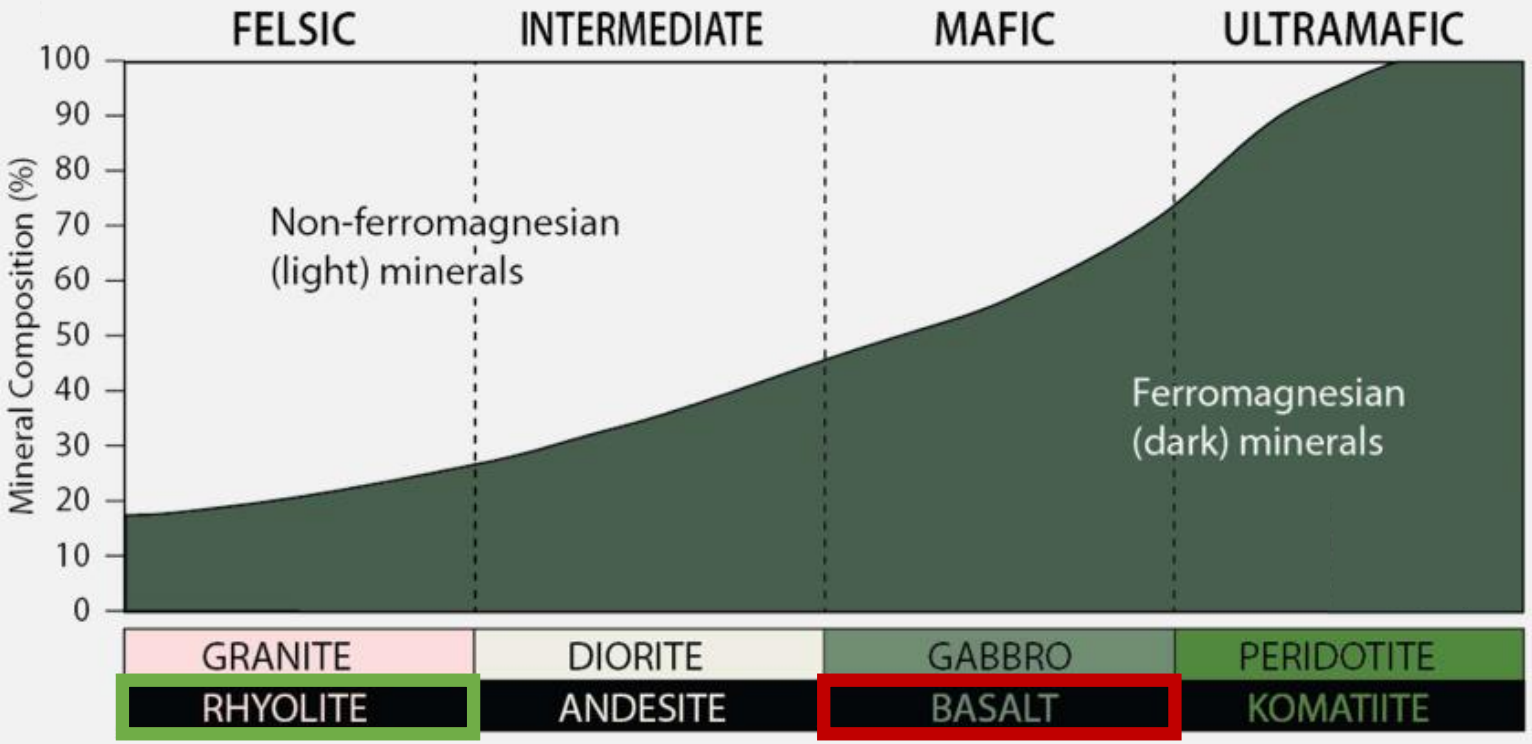


holomelanocratic
pyroxenite

Gill (2010) Fig. 1.3


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# Colour is correlated with magma composition






basalt: mafic



obsidian: felsic

Beware – fine-grained rocks are often unusually dark for their composition


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# Igneous processes: Crystallization

2.1 Describing igneous hand specimens

**2.2 Mineralogical classification of igneous rocks**





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# Mineralogical classification: gathering information

More information in the Practical Reference Handout

- Grain size? Size of groundmass crystals dictates whether to use coarse-grained or fine-grained rock name
- What minerals? In what proportions? Describe minerals first, then identify.
- Are all the minerals primary (crystallized from the magma)? Any secondary minerals (formed due to alteration of the rock)?

| Mineral, modal abundance | Size, habit, shape | Relief, colour, pleochroism, cleavage | Interference colour, type of extinction | Twinning, zoning, exsolution | Alteration? | Textural association* | ID |
|---|---|---|---|---|---|---|---|
| A<br>40% | 1-2 mm phenocrysts, euhedral tabular.<br><0.1 mm groundmass, euhedral acicular. | Low relief, colourless.<br>No cleavage. | Low first-order. Inclined extinction with respect to crystal edges. | Multiple lamellar twinning.<br>Large phenocrysts show concentric zoning.<br>No exsolution. | Some cloudy alteration in cores of largest crystals. | | Plagioclase |
| B<br>40% | <0.1 mm, found in groundmass only | Medium relief, faint brown | … | … | | | |
| C *etc.* | | | | | | | |


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# A simple mineralogical classification method

More information in the Practical Reference Handout

| Root name | Essential minerals required | Possible type minerals |
|---|---|---|
| basalt<br>dolerite/gabbro | augite + calcic plag | olivine, LCP (→ tholeiitic basalt ),<br>minor nepheline (+ olivine → alkali basalt ) |
| andesite<br>diorite | sodic plag[2] + a mafic mineral (pyroxene *or* hornblende *or* biotite) | pyroxene, hornblende or biotite (according to the dominant mafic mineral present) |
| dacite<br>granodiorite | sodic plag + alkali feldspar + quartz (>20%) | hornblende, biotite |
| rhyolite<br>granite | alkali feldspar ± sodic plag + quartz (>20%) | biotite, aegirine |

Gill (2010) Table 1.1

LCP = low-Ca pyroxene (e.g. enstatite, pigeonite)

- Essential minerals are implied by the root name. Type minerals are not implied by the root name.
- Essential and type minerals together are the major minerals of the rock.
- Accessory minerals (not listed in the table) comprise only a very small proportion of the rock and are not included in the rock name.


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# Simple mineralogical classification: Worked example 1



Grain size:

- Phenocrysts of various sizes
- Fine-grained groundmass

Need fine-grained root name

Minerals:

- Olivine

Type mineral for basalt

- Clinopyroxene
- Plagioclase (calcic)

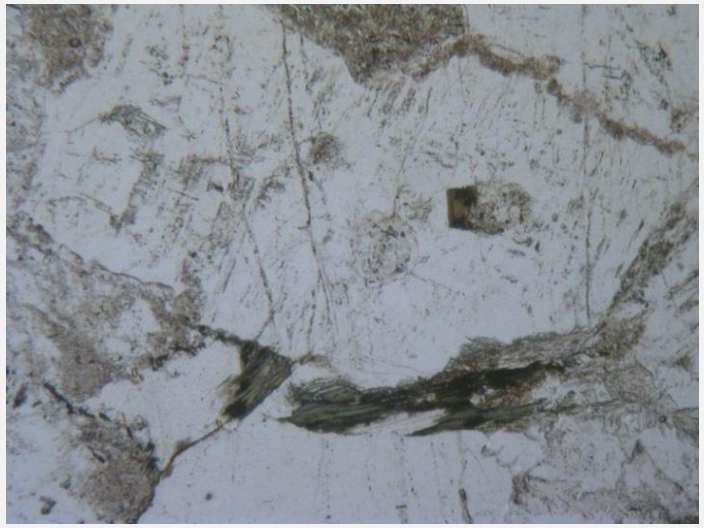
Essential minerals for basalt

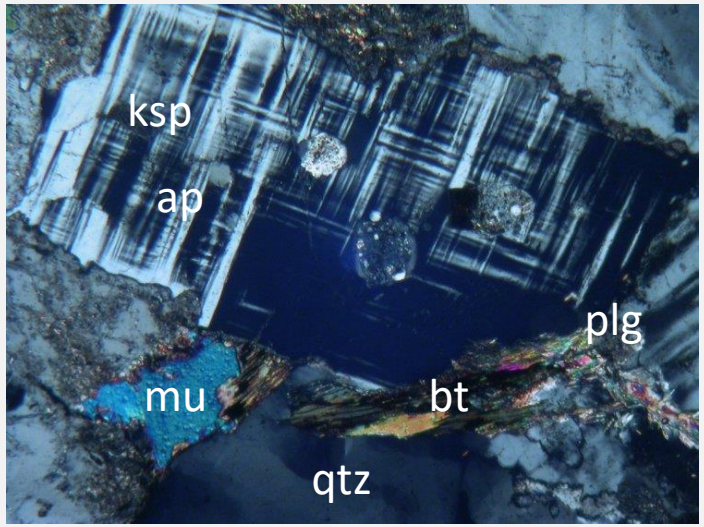
olivine basalt



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# Simple mineralogical classification: Worked example 2






Grain size:

- Coarse

Need coarse-grained root name

Minerals:

- Alkali feldspar (microcline)
- Plagioclase (sodic)
- Quartz

Essential minerals for granite and granodiorite

- Muscovite
- Biotite

Type minerals for granite (to be discussed in Week 9)

- Apatite

Accessory mineral – not important for rock name

muscovite > biotite

**biotite muscovite granite**

If there is more than one type mineral, list the least abundant first.

It is unusual to see more than two type minerals listed in the rock name.



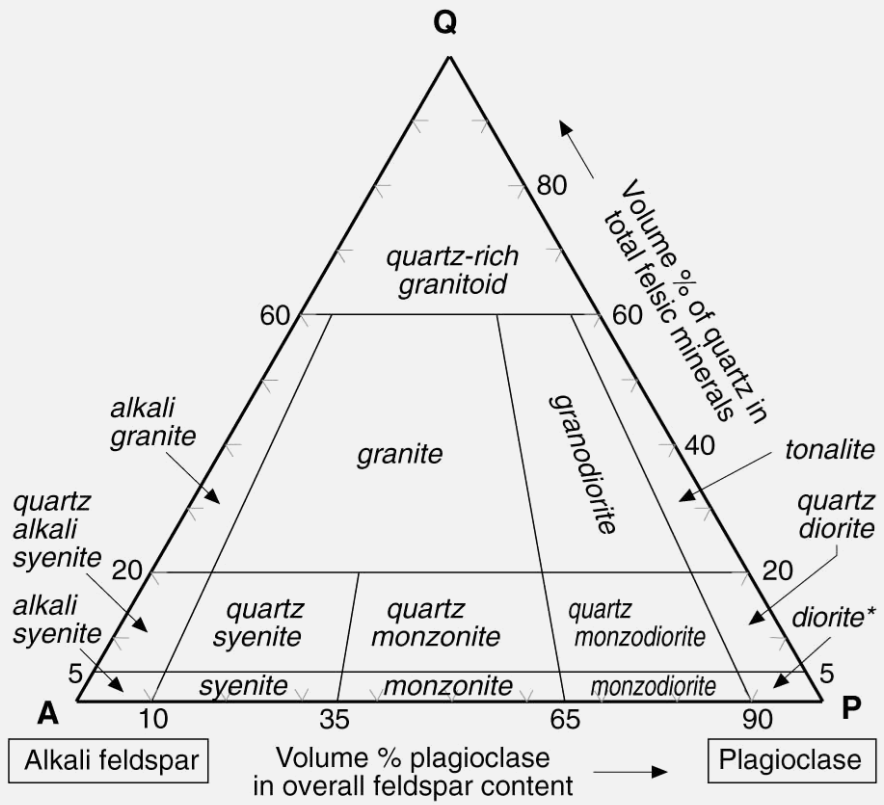
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# Quantitative classification schemes for coarse-grained rocks

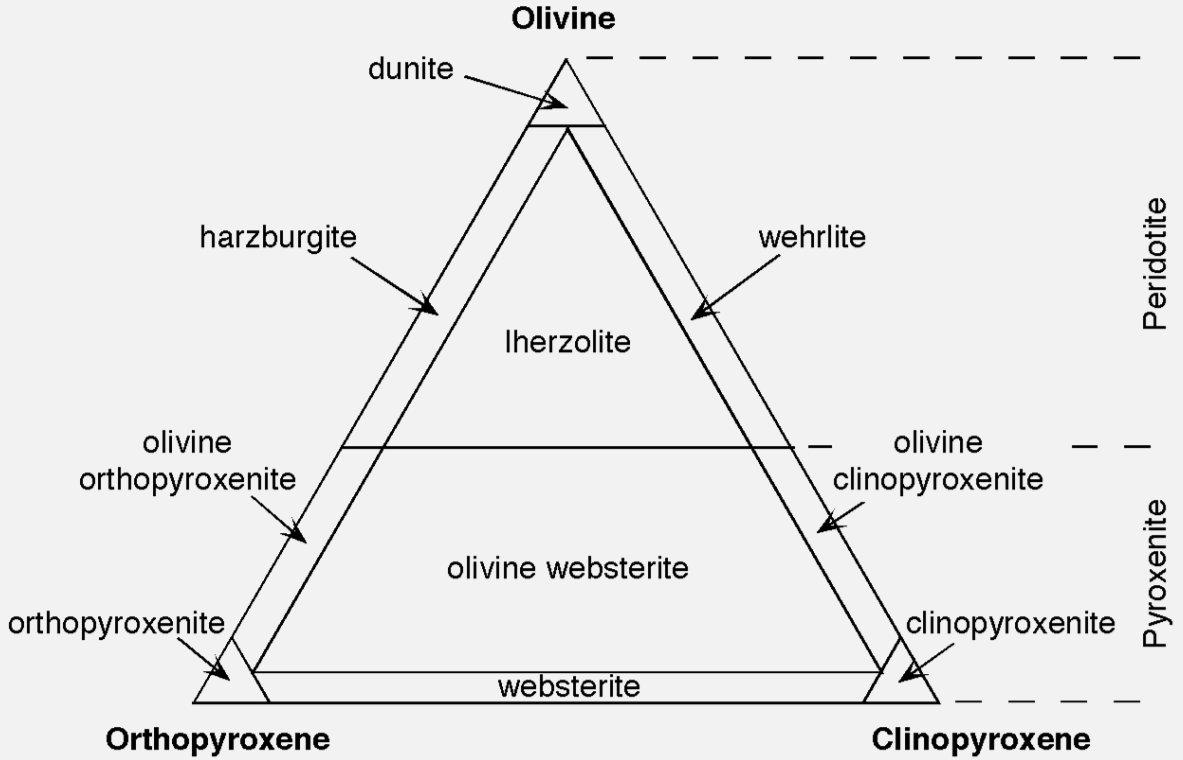
Guide to reading and plotting triangular diagrams provided on Blackboard, in Week 4 folder

## Granitoid rocks: QAP(F) diagram - Week 9

(quartz-alkali feldspar-plagioclase-feldspathoid)




## Ultramafic rocks - Week 5





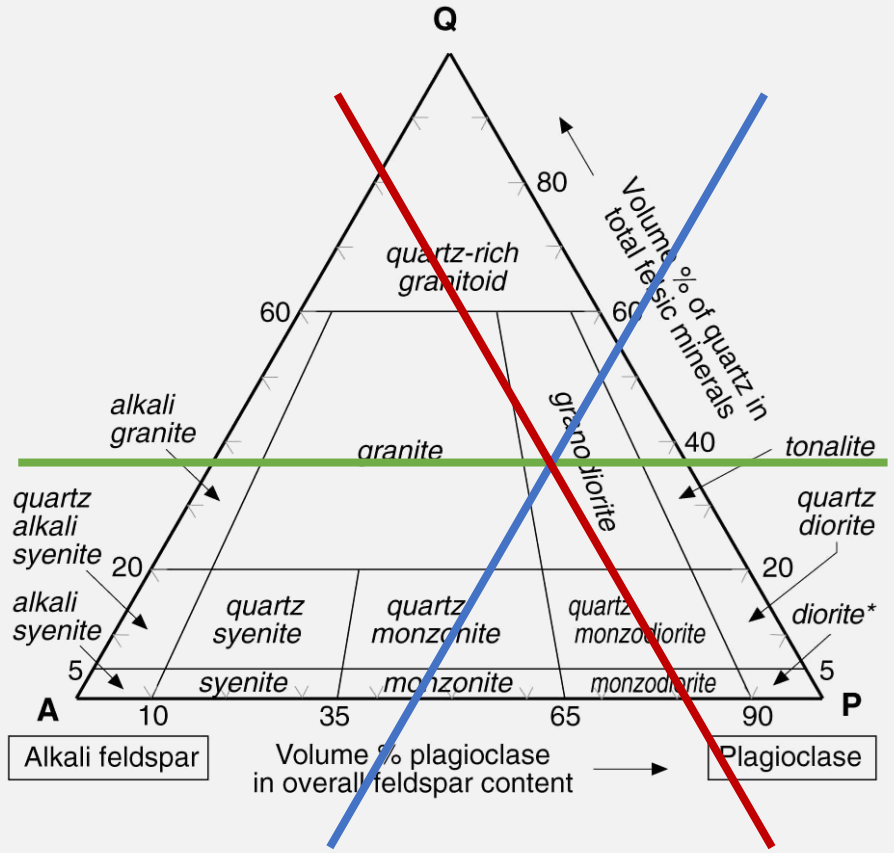

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# Quantitative classification: Worked example 3

Guide to reading and plotting triangular diagrams in Week 4 Blackboard folder

## Granitoid rocks: QAP(F) diagram

(quartz-alkali feldspar-plagioclase-feldspathoid)





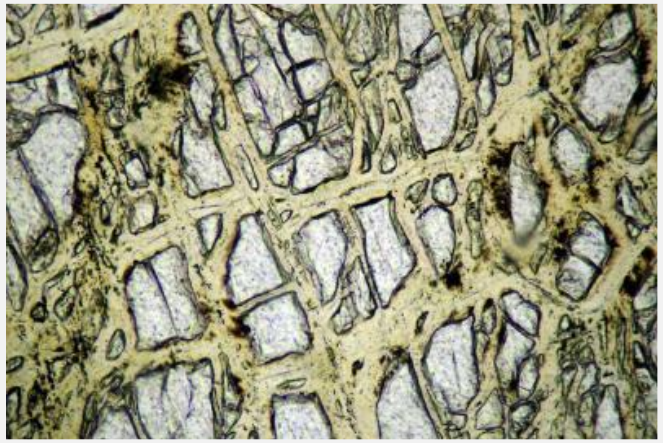

| | |
|---|---|
| 19% | 15% alkali feldspar (orthoclase) |
| 44% | 35% plagioclase |
| 37% | 30% quartz |
| | 20% biotite |

biotite granodiorite



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# Secondary minerals



serpentine replacing olivine



sericite (clay mineral) replacing plagioclase

Secondary minerals form after the magma has completely crystallized. They may fill void spaces or may replace the original minerals.



chlorite replacing biotite



amygdaloidal basalt



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# Igneous processes: Crystallization

Familiarize yourself with the main classification schemes – all in the Practical Reference Handout

You should now be able to:

- Describe the grain size and colour index of igneous rocks
- State the essential minerals found in common igneous rock types: basalt/gabbro, andesite/diorite, and rhyolite/granite
- Explain the difference between primary and secondary minerals
- Classify an igneous hand specimen or thin section on the basis of its mineralogy





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# Igneous processes: Crystallization

**3.1 Crystal nucleation and growth**

3.2 Igneous textures and the order of crystallization

3.3 Crystal size distributions

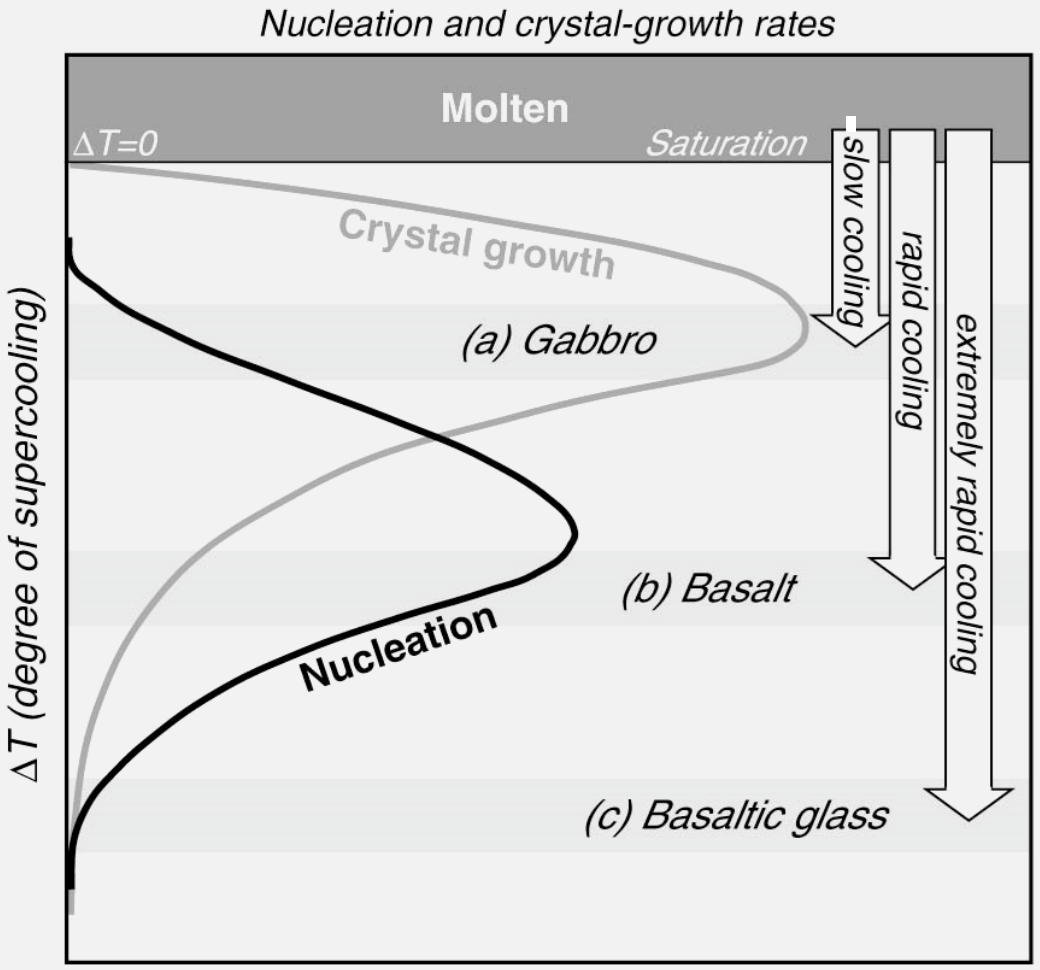




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# Nucleation and growth

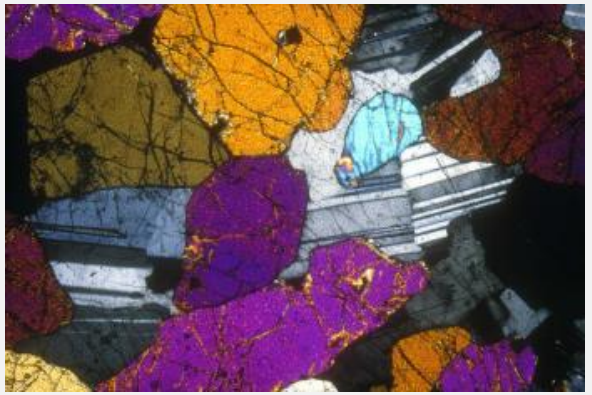
Beware! – same notation used for G (crystal growth) and G (Gibbs free energy). It should be clear from the context which is meant.




Gill (2010) Fig. 2.3.1

Grain size is proportional to the ratio of the **crystal growth rate** (G) to the **nucleation rate** (N)





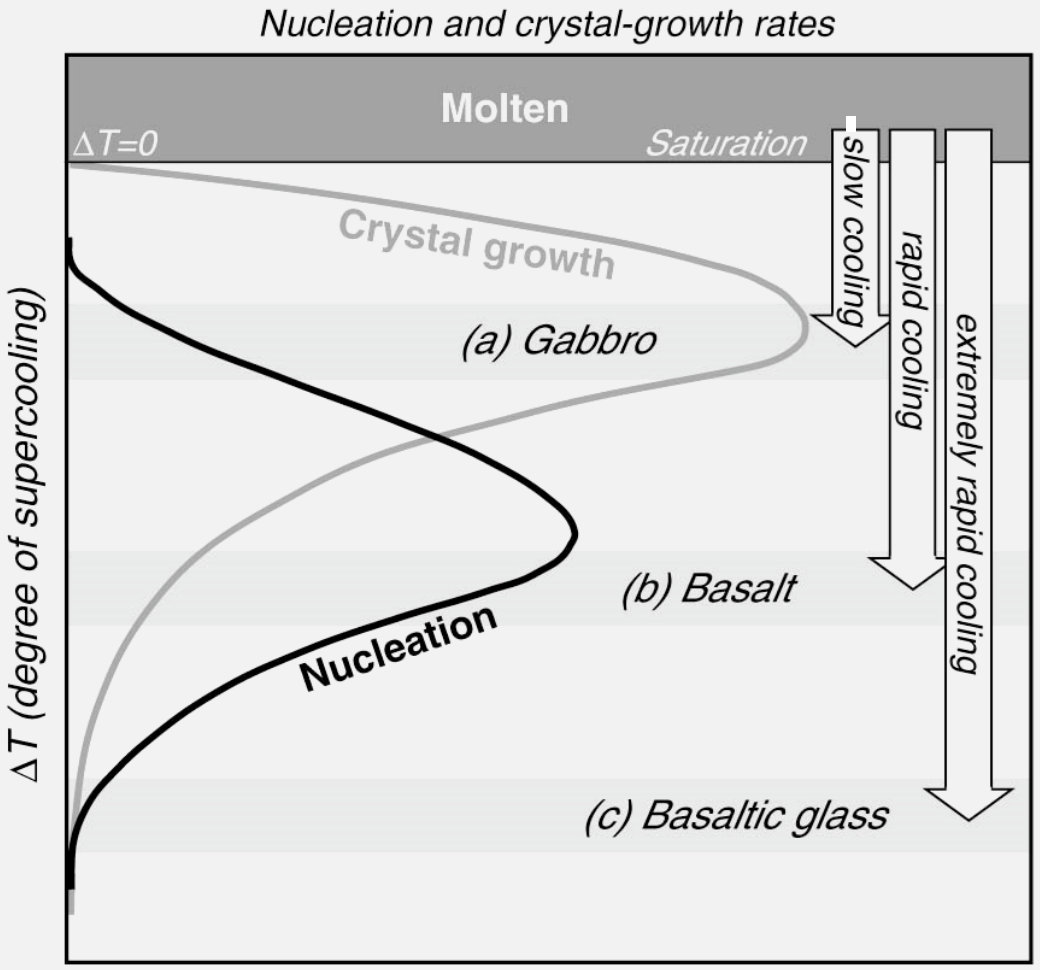
Plutonic settings where heat loss is slow:

- G >> N
- coarse-grained textures



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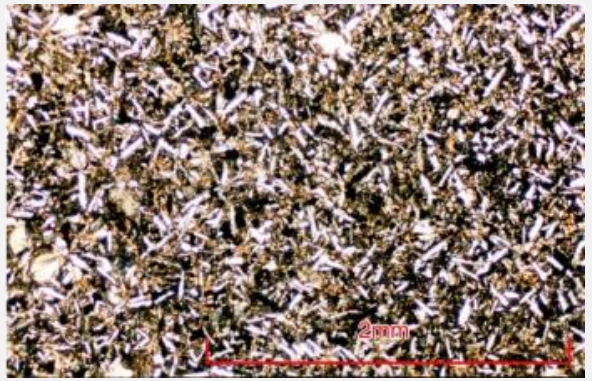
# Nucleation and growth




Gill (2010) Fig. 2.3.1

Grain size is proportional to the ratio of the **crystal growth rate** (G) to the **nucleation rate** (N)



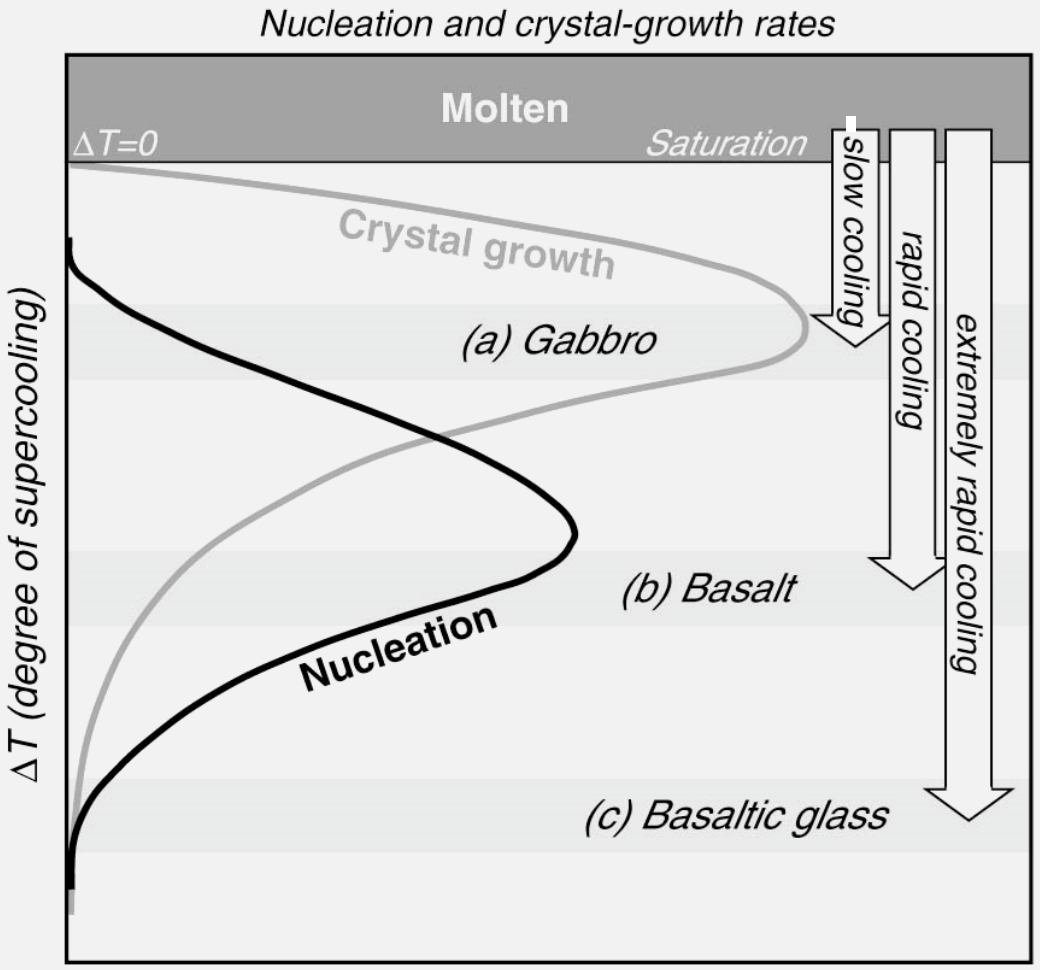



Volcanic settings where heat loss is fast:

- N >> G
- fine-grained textures

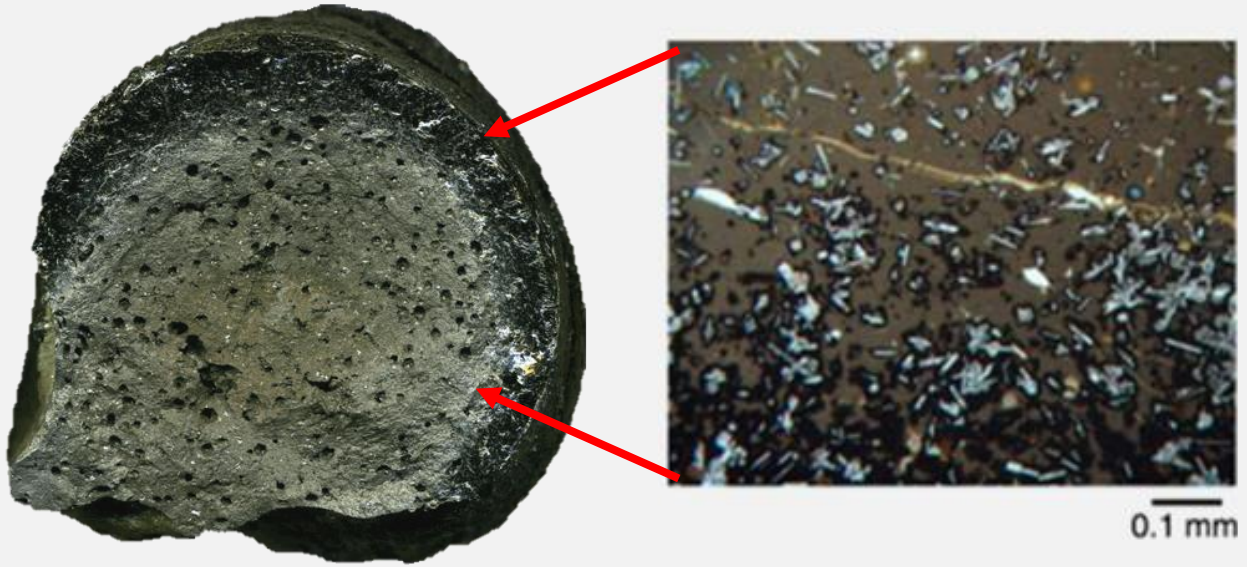

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# Nucleation and growth




Gill (2010) Fig. 2.3.1

Grain size is proportional to the ratio of the **crystal growth rate** (G) to the **nucleation rate** (N)

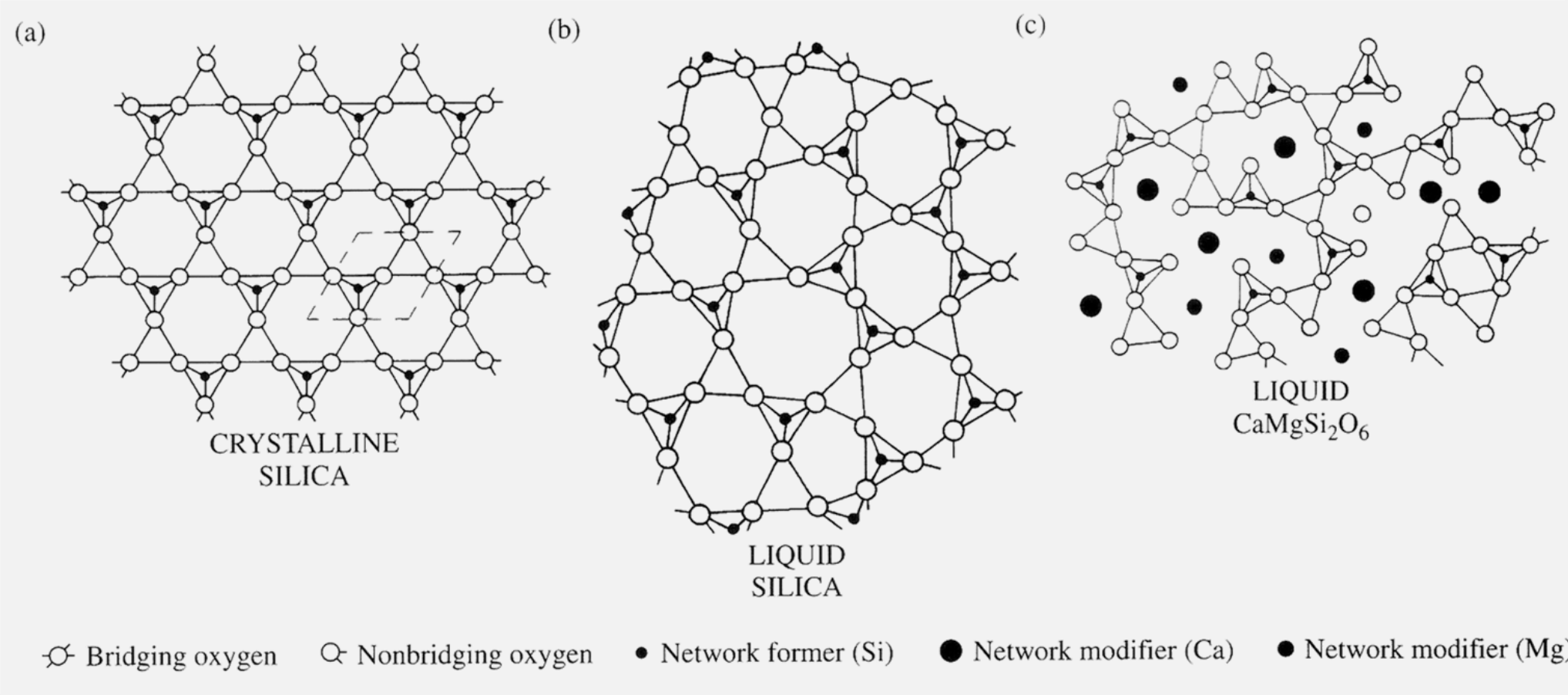



Very fast cooling or instantaneous quenching:

- N and G ≈ 0
- Glass


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# Nucleation and growth require cations to be able to diffuse through the silicate melt

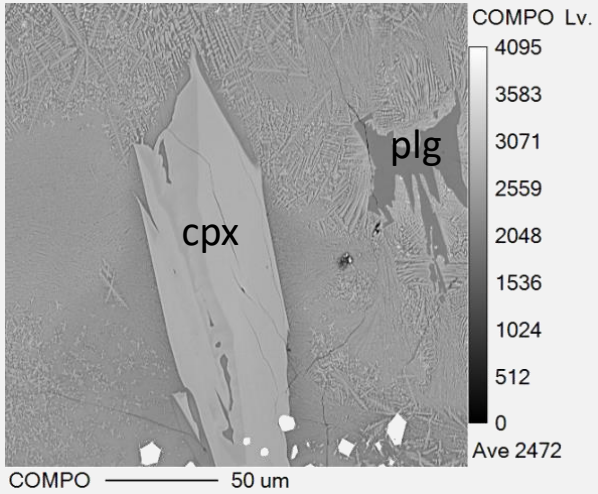


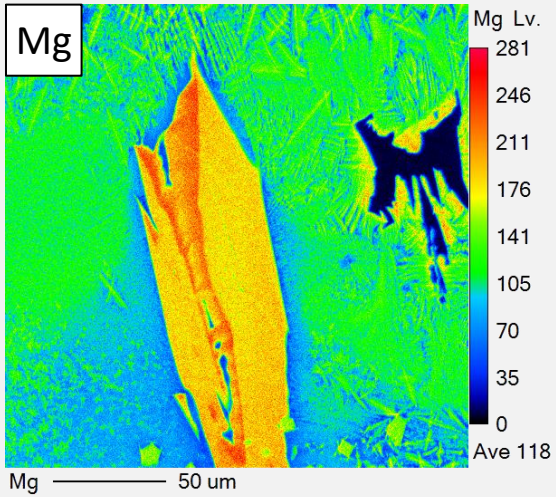

Best (2006) Fig. 4.2

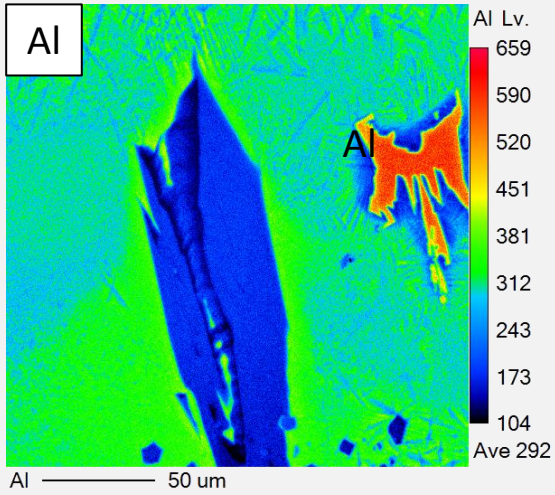
It is easier for cations to diffuse through low-silica mafic magmas than through high-silica felsic magmas. So nucleation and growth are easier in basalts than in rhyolites.

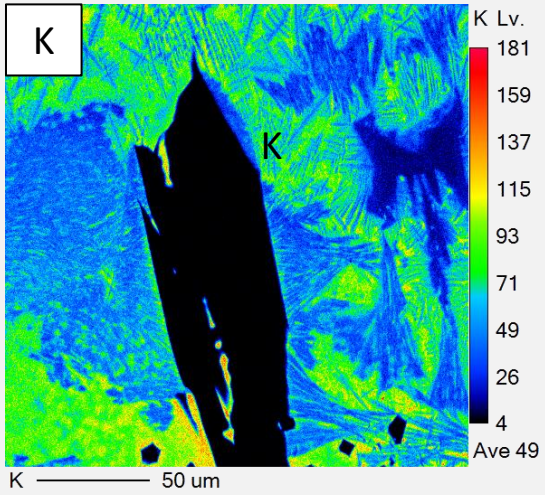

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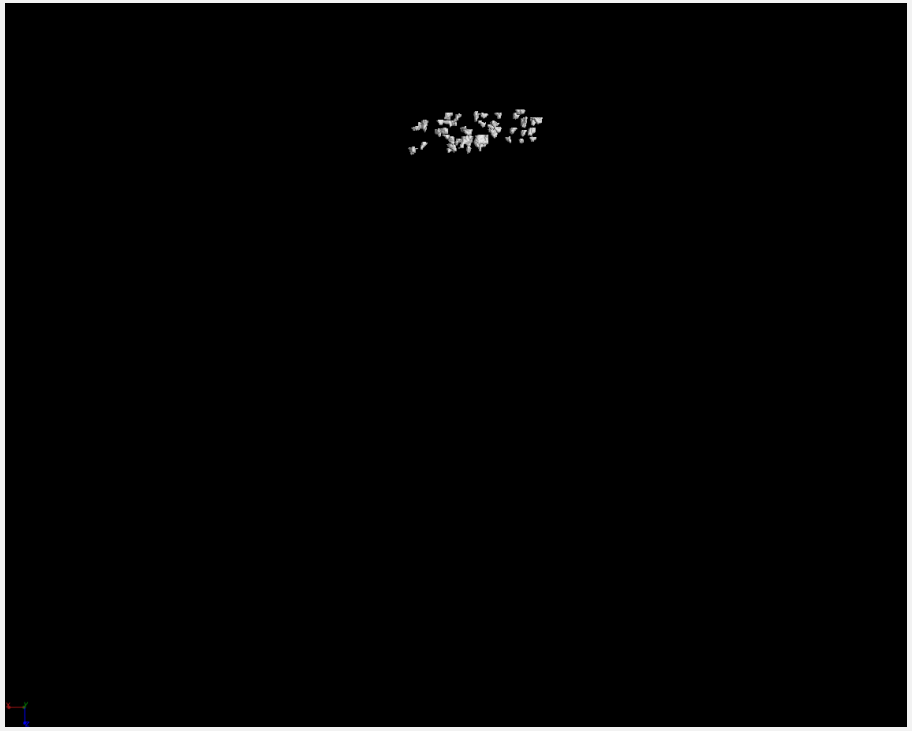
# Crystal nucleation and growth in basalts



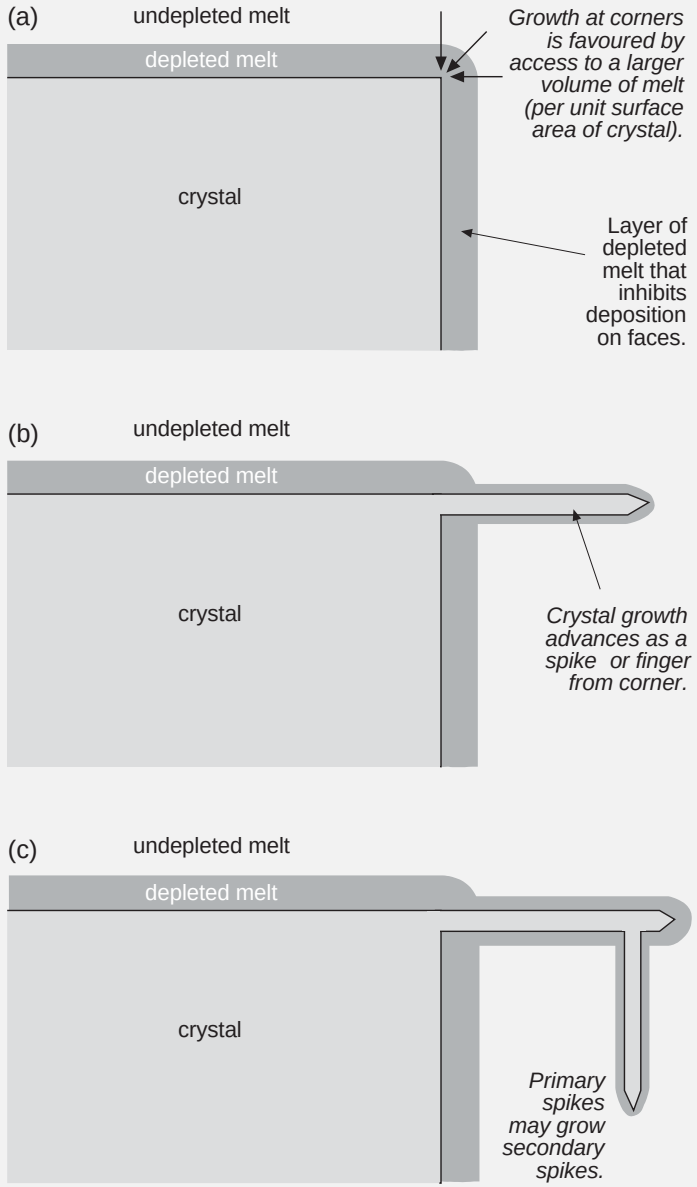



Polacci et al. (2018)
https://doi.org/10.1038/s41598-018-26644-6

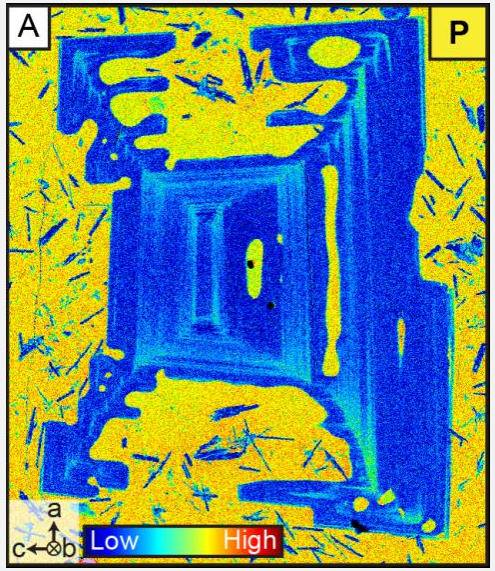

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Gill (2010) Fig. 5.10



Quench olivine textures in Apollo 12 mare basalts




Phosphorus chemical zoning in olivine from Kilauea, Hawaii

Shea et al. (2015)
https://doi.org/10.1130/G37082.1


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# Preventing crystallization: ultra-rapid quenching of magma to form glass

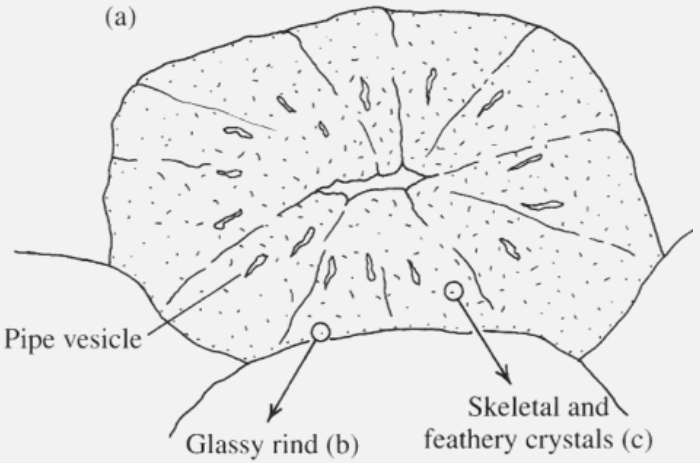
## Basalt glass: subaerial eruptions





## Basalt glass: subaqueous eruptions











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# Igneous processes: Crystallization

3.1 Crystal nucleation and growth

**3.2 Igneous textures and the order of crystallization**

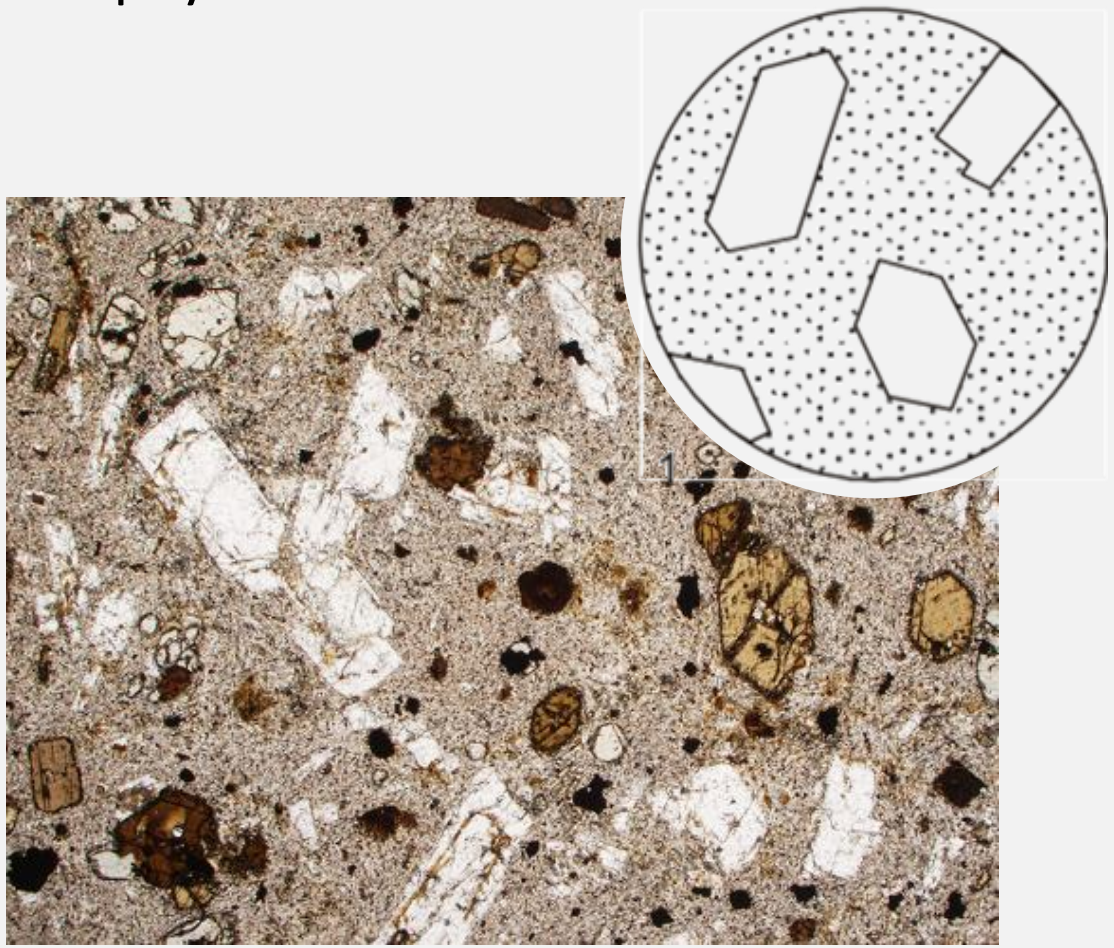
3.3 Crystal size distributions





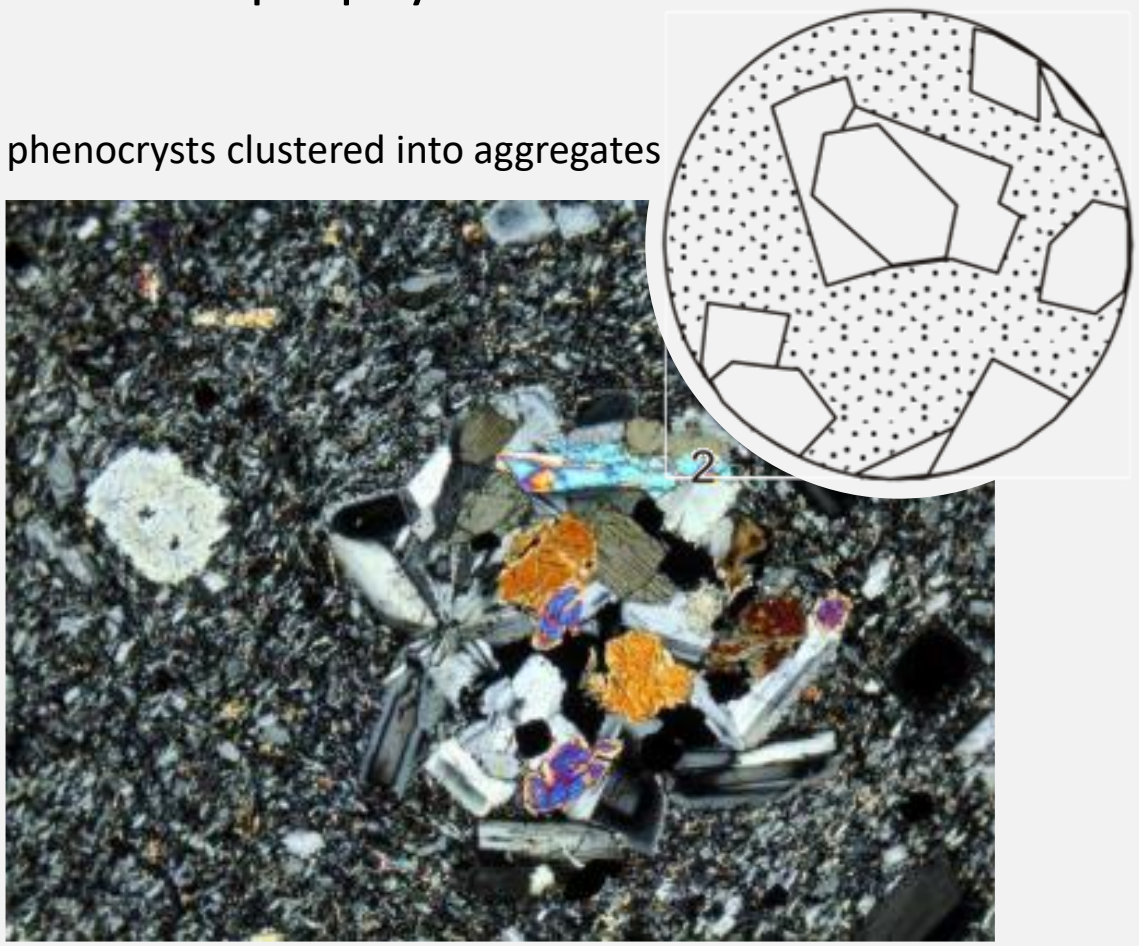
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## Porphyritic texture



## Glomeroporphyritic texture

phenocrysts clustered into aggregates



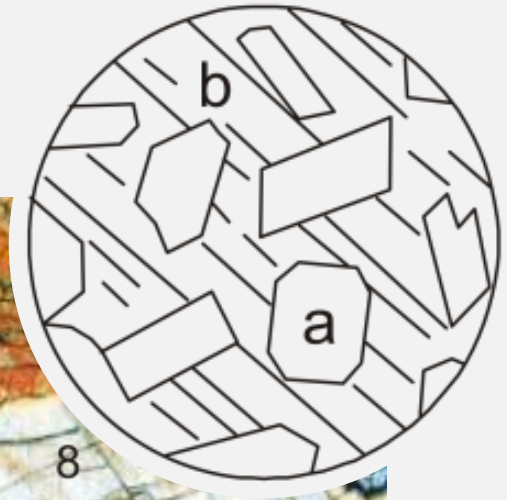
Phenocrysts grew first, slowly, in a magma chamber. Larger phenocrysts had longer to grow than smaller phenocrysts.

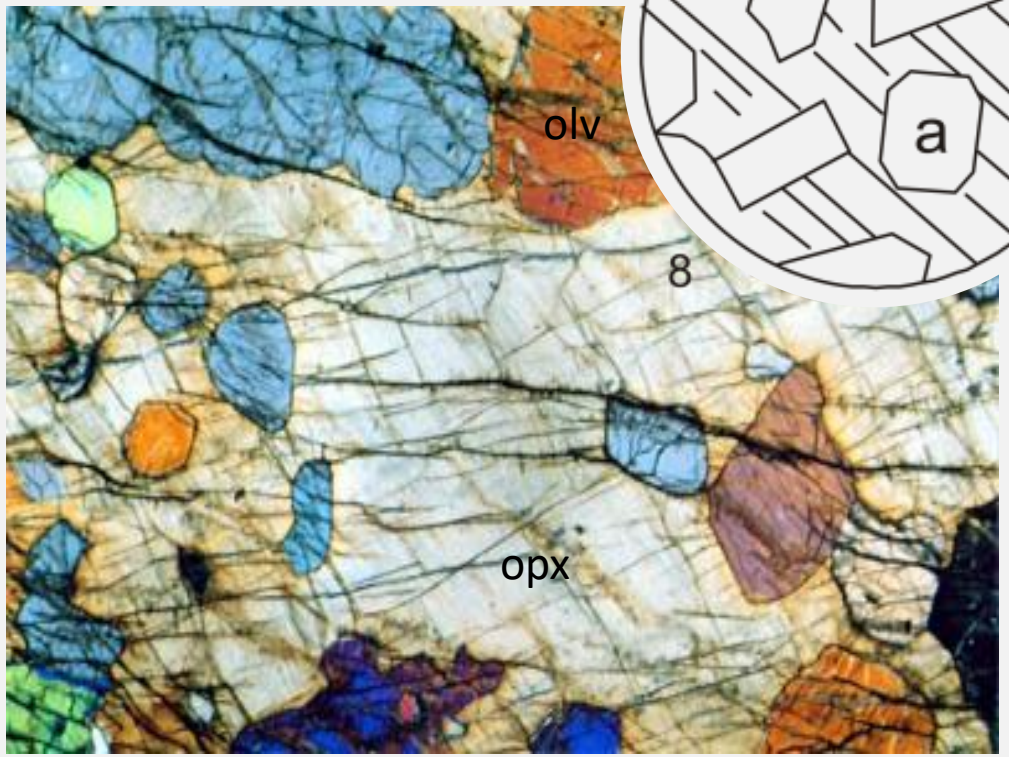
Groundmass crystals grew last, very quickly, when the magma was erupted to the surface.



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# Poikilitic texture




Many crystals of mineral A are fully enclosed by one large crystal of mineral B.

Mineral A = chadacrysts (means “contained crystals”)

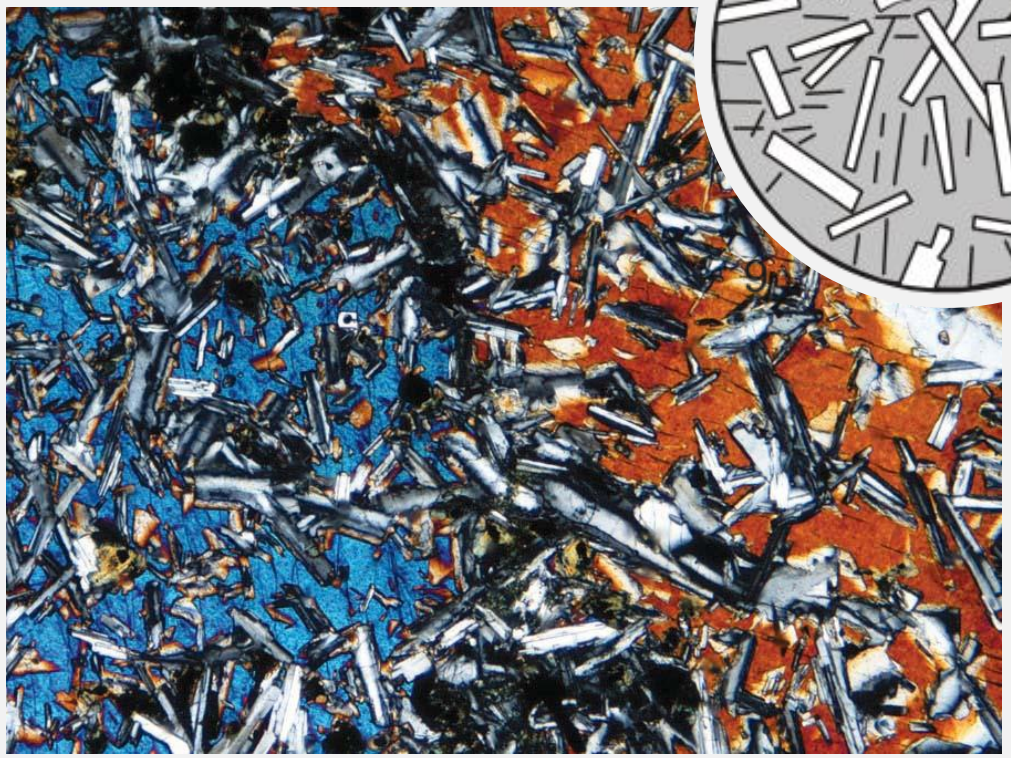
Mineral B = oikocryst (means “house crystal”)

Order of crystallization:
1. Mineral A
2. Mineral B


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# Ophitic texture

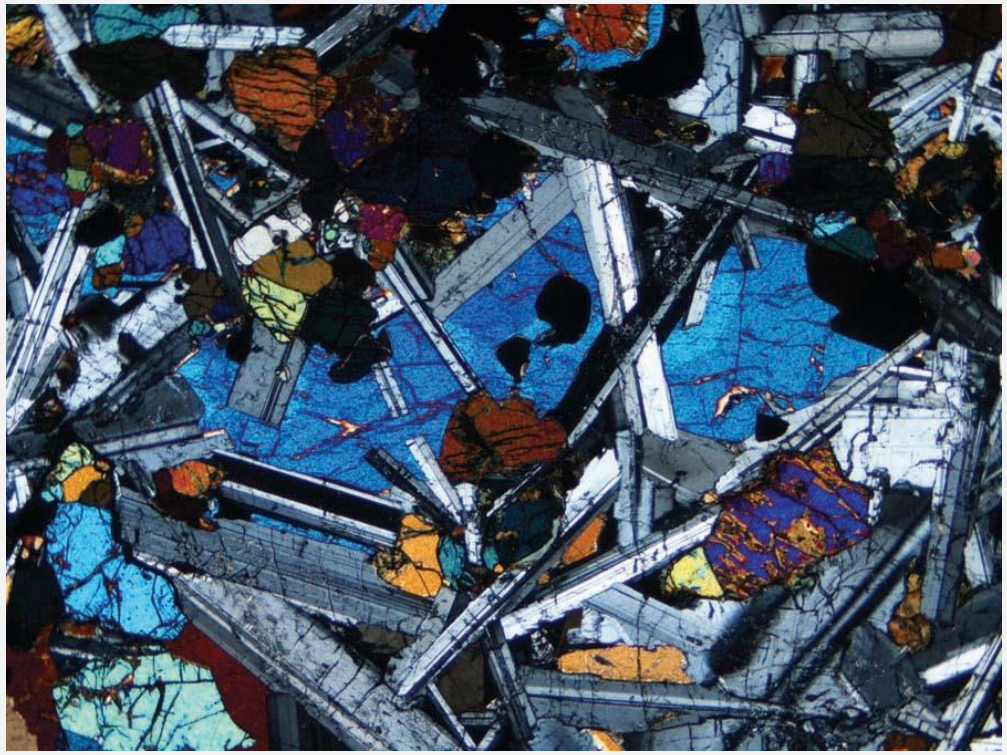
Special case of poikilitic texture involving plagioclase and clinopyroxene





Randomly oriented plagioclase laths fully enclosed by large clinopyroxenes (augite)

# Sub-ophitic texture

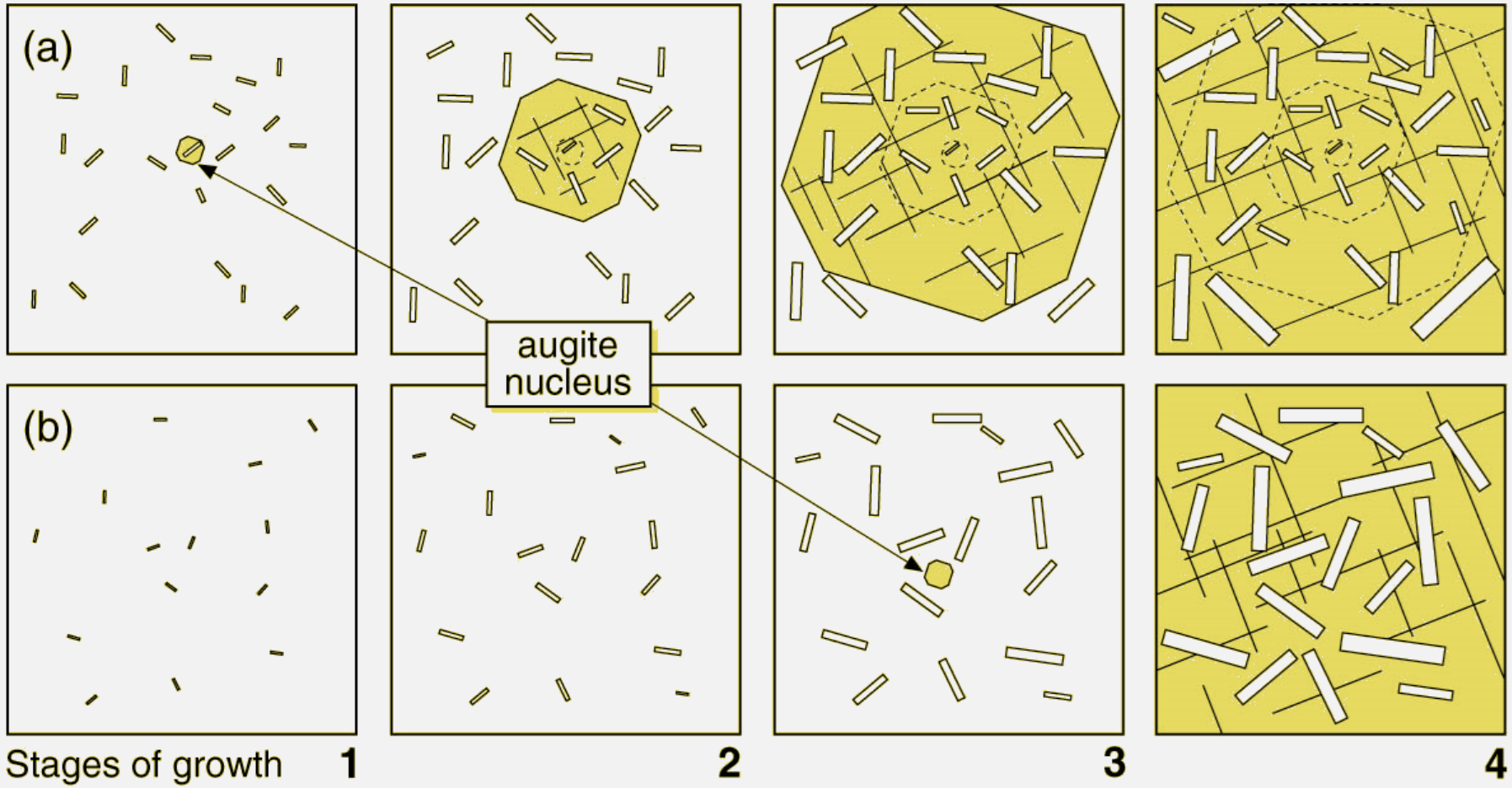


Plagioclase laths partially enclosed by clinopyroxene (augite)



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# Two ways that ophitic textures can form




(a) Plag and cpx nucleate at the same time, but cpx grows faster and encloses the slow-growing plagioclases.

Order of crystallization:
1. Plag + cpx

(b) Plag nucleates first. Cpx nucleates second. Cpx grows faster and encloses the slow-growing plagioclases.

Order of crystallization:
1. Plag
2. Plag + cpx

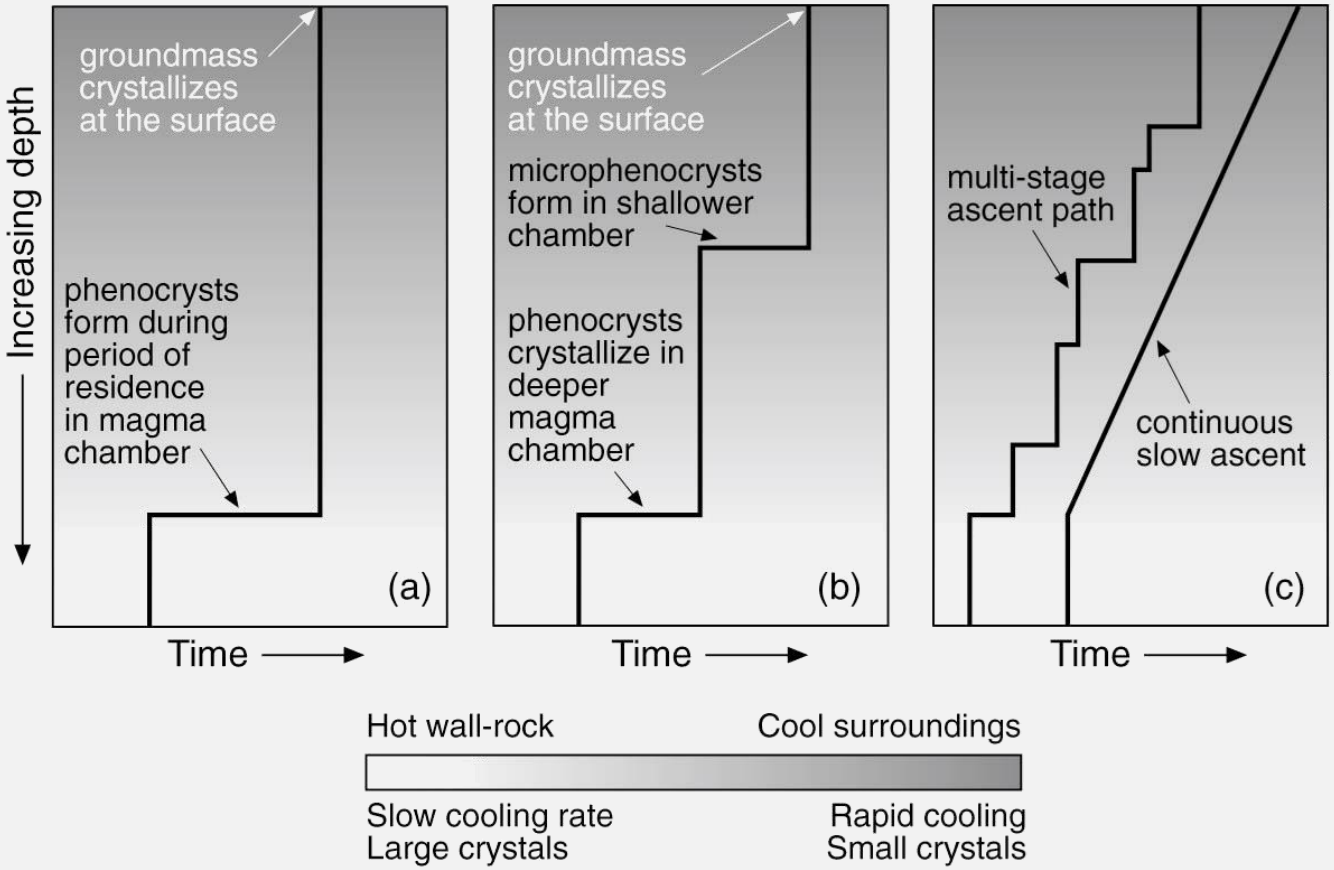

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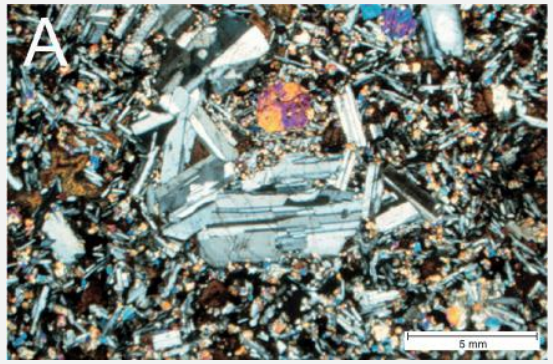
# Seriate texture: gradual and continuous range in crystal size

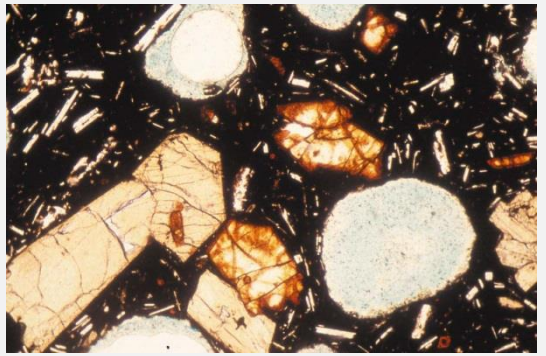
Porphyritic: phenocrysts + groundmass

Porphyritic: phenocrysts + microphenocrysts + groundmass

Seriate: continuous range of crystal sizes



Order of crystallization: Largest phenocrysts likely to have grown first. Groundmass crystals grew last.

Interpretation: Multiple episodes of nucleation as the magma ascends through stacked magma chambers in the crust.

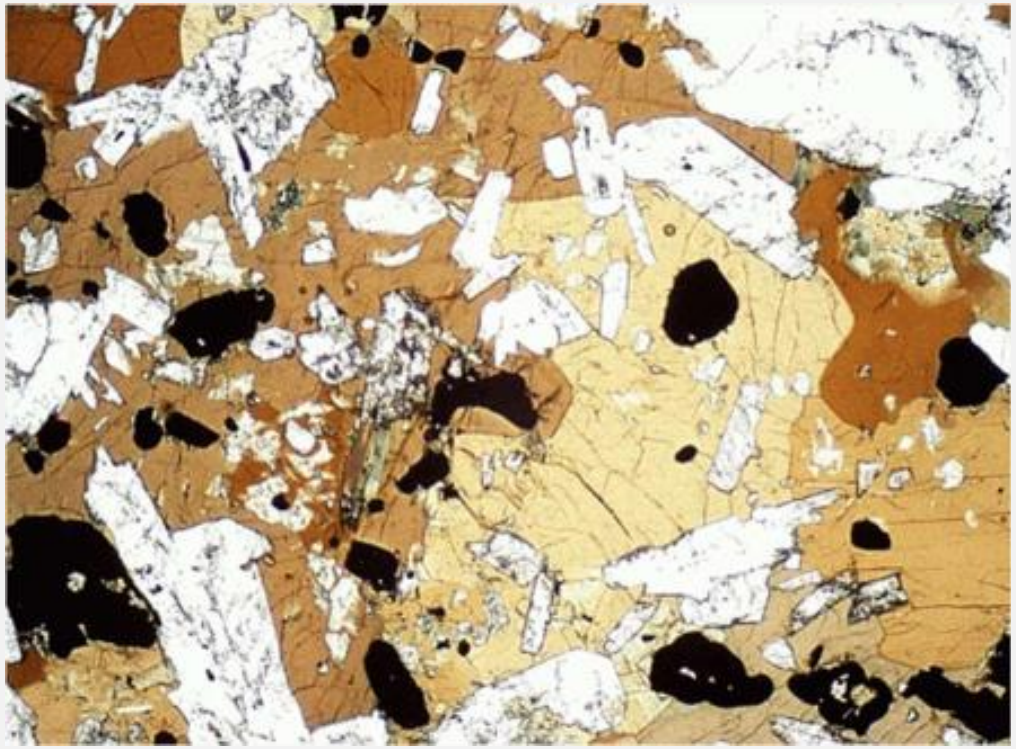


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# Mineral inclusions provide evidence for order of crystallization



Apatite inclusions in augite



Magnetite inclusions in hornblende

Order of crystallization: The enclosed mineral (inclusion) started crystallizing first. The surrounding mineral grew later.


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# Igneous processes: Crystallization

3.1 Crystal nucleation and growth

3.2 Igneous textures and the order of crystallization

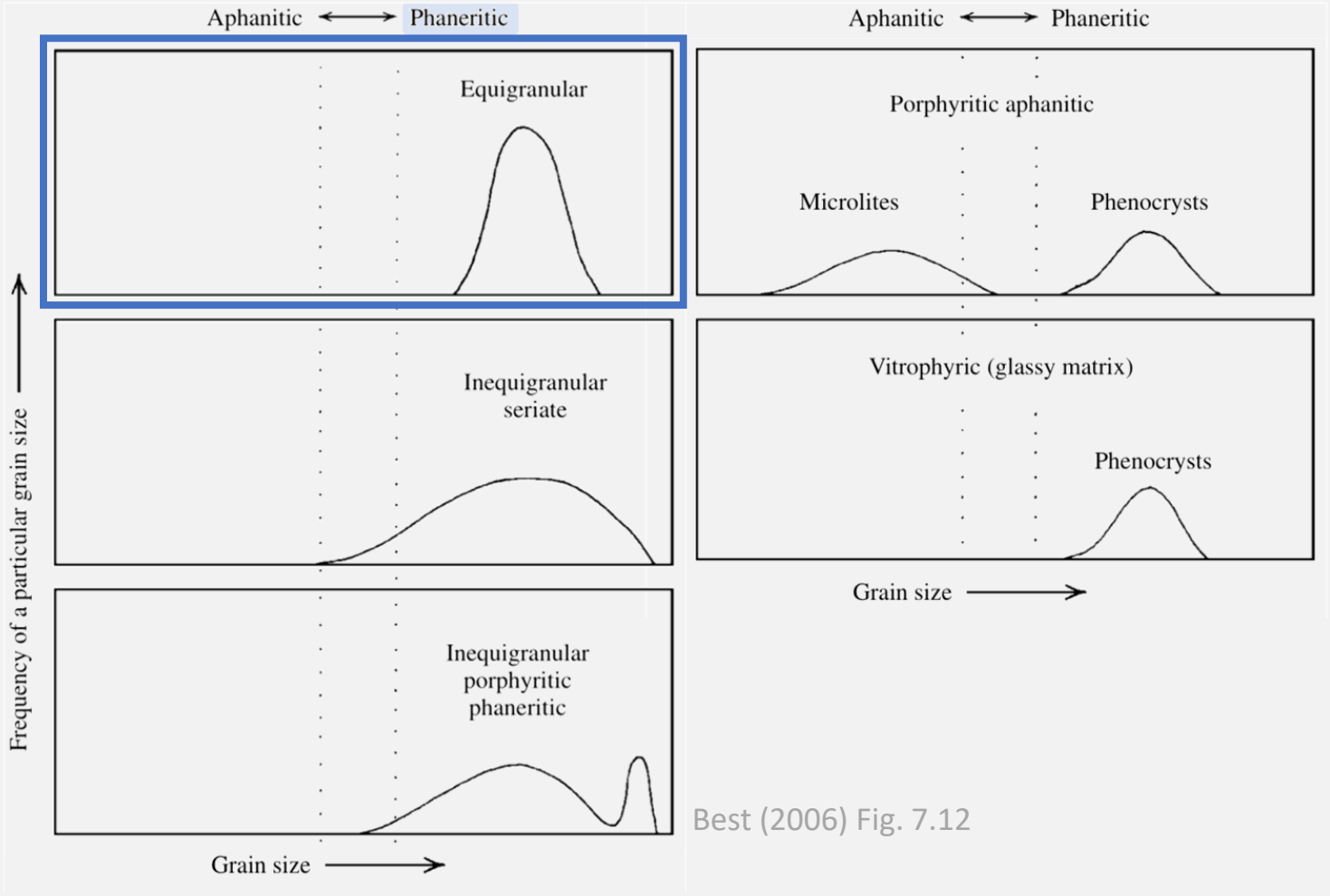
**3.3 Crystal size distributions**



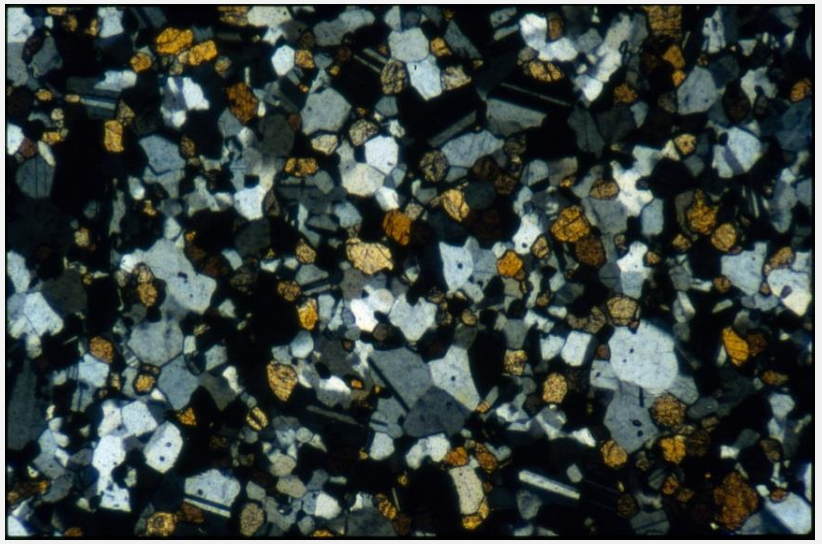


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# Crystal size distributions




Best (2006) Fig. 7.12

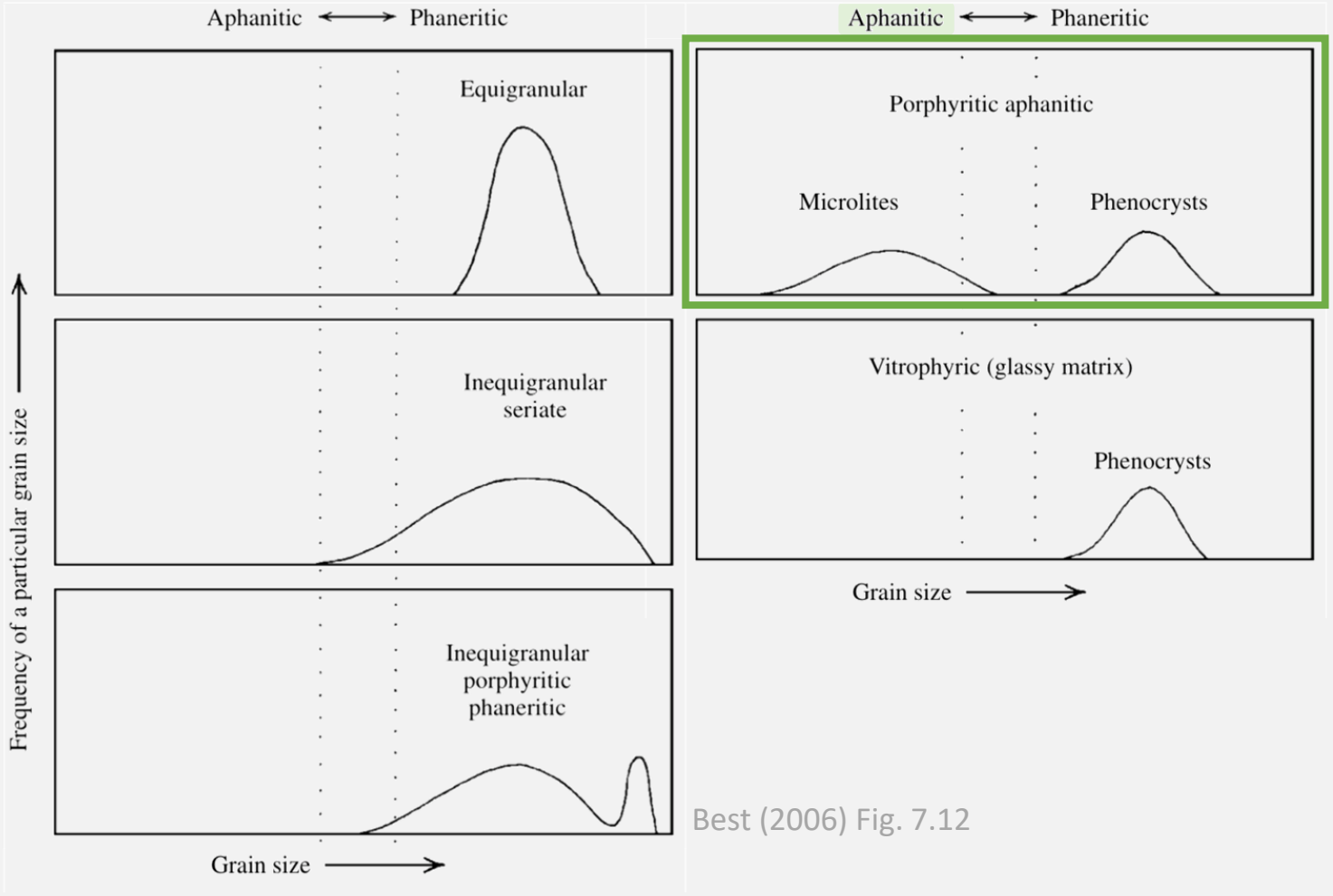


Equigranular gabbro



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# Crystal size distributions




Best (2006) Fig. 7.12



Porphyritic aphanitic basalt


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# Igneous processes: Crystallization

Familiarize yourself with important igneous textures – all in the Practical Reference Handout

You should now be able to:

- Explain the relationship between cooling rate, nucleation rate and growth rate
- Describe and interpret igneous textures using observations from thin sections
- Interpret the magma cooling rate using observations of crystal size and crystal shape
- Use textural observations to deduce information about the order of crystallization




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# Igneous processes: Crystallization

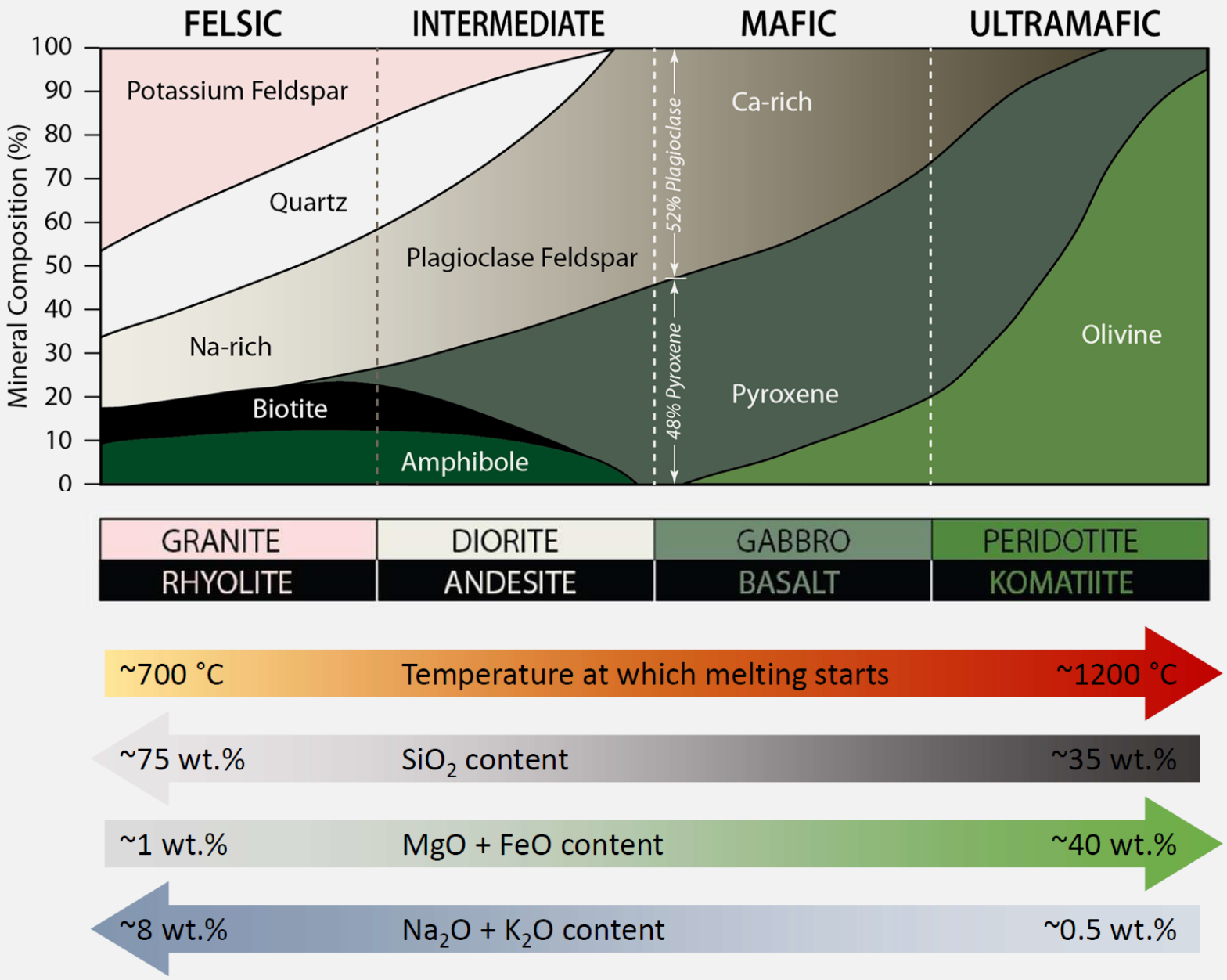
**3.1 Chemical classification of igneous rocks**

3.2 Liquid lines of descent





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## Mineralogical classification:

- Quick, easy and cheap: only needs a hand specimen
- Can be done in the field
- Thin section analysis needs only a microscope

But:

- Not good for fine-grained rocks
- Limited compositional information

## Chemical classification:

- Quantitative compositional data
- Works for any grain size

But:

- Expensive: requires laboratory analysis
- Time-consuming sample preparation
- No mineralogical information


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# Major element compositions of common igneous rocks

See also EART 26201 Principles of Geochemistry, Lecture 1

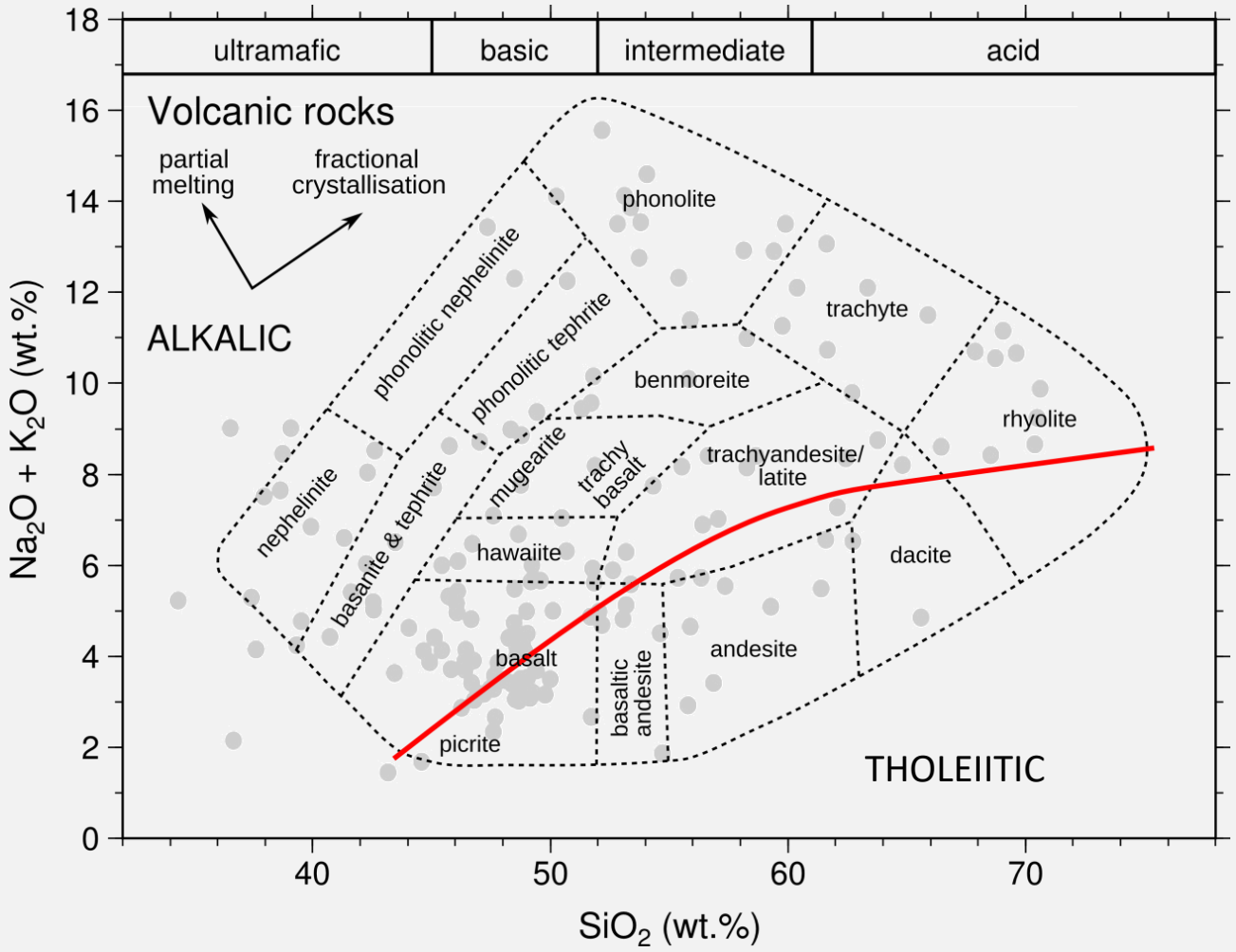
| wt.% | Basalt 1 | Basalt 2 | Andesite | Dacite | Rhyolite |
|---|---|---|---|---|---|
| $SiO_2$ | 48.32 | 49.62 | 56.31 | 64.89 | 70.28 |
| $TiO_2$ | 0.93 | 1.66 | 0.89 | 0.32 | 0.24 |
| $Al_2O_3$ | 15.99 | 14.23 | 16.41 | 15.3 | 13.88 |
| FeO | 8.96 | | | 3.34 | 3.52 |
| $Fe_2O_3$ | | 13.22 | 7.37 | 0.73 | 0.85 |
| MnO | 0.16 | 0.21 | 0.12 | 0.08 | 0.14 |
| MgO | 10.56 | 7.19 | 4.66 | 2.40 | 0.06 |
| CaO | 14.01 | 11.94 | 6.59 | 4.69 | 0.44 |
| $Na_2O$ | 1.72 | 2.10 | 3.96 | 4.34 | 5.27 |
| $K_2O$ | 0.05 | 0.19 | 1.41 | 1.71 | 5.24 |
| $P_2O_5$ | 0.01 | 0.16 | 0.24 | 0.07 | 0.01 |
| $H_2O$ | | 0.73 | 0.20 | | |
| Total | 100.69 | 99.78 | 97.96 | 99.37 | 99.80 |

As crystallization proceeds:

- Compatible elements are sequestered into crystallizing minerals
- Incompatible elements become concentrated in the melt


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# Total alkalis vs. silica (TAS) diagram






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# Igneous processes: Crystallization

3.1 Chemical classification of igneous rocks

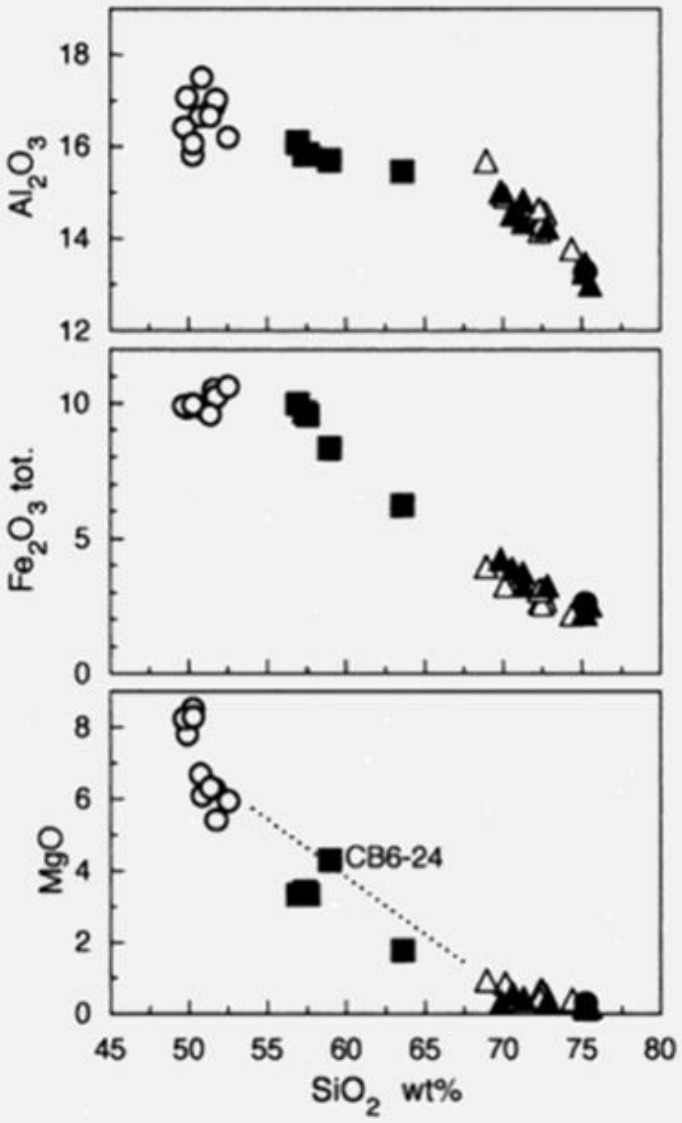
**3.2 Liquid lines of descent**

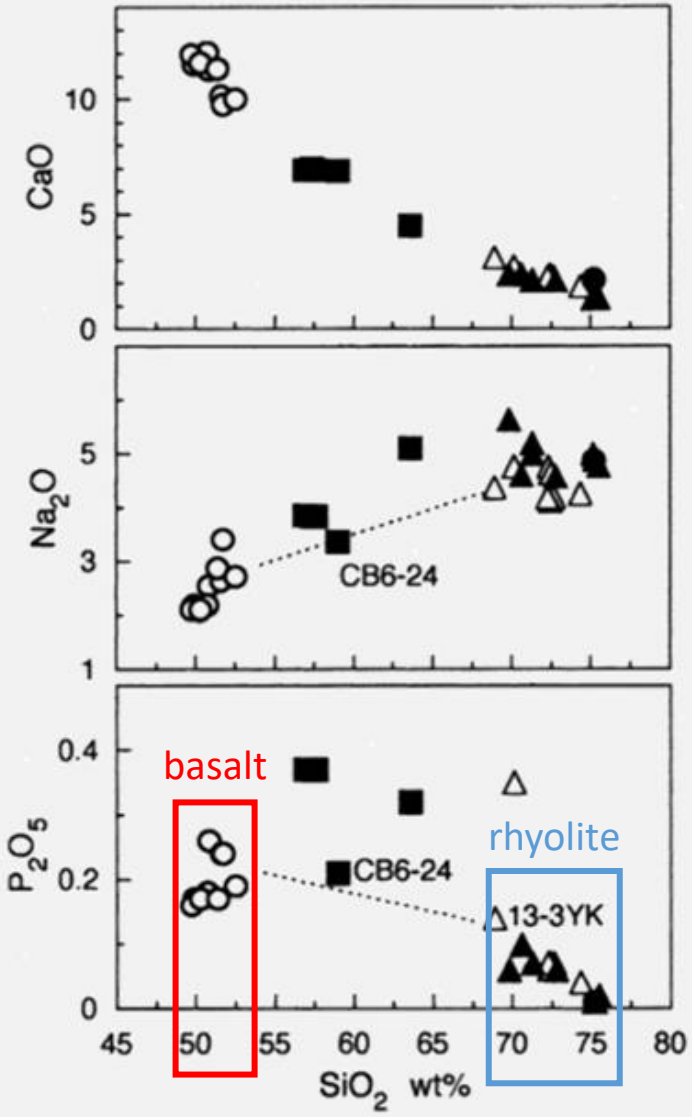




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# Variation diagrams




Variation diagrams show how the abundances of major (or trace) elements vary within a suite of igneous rocks.

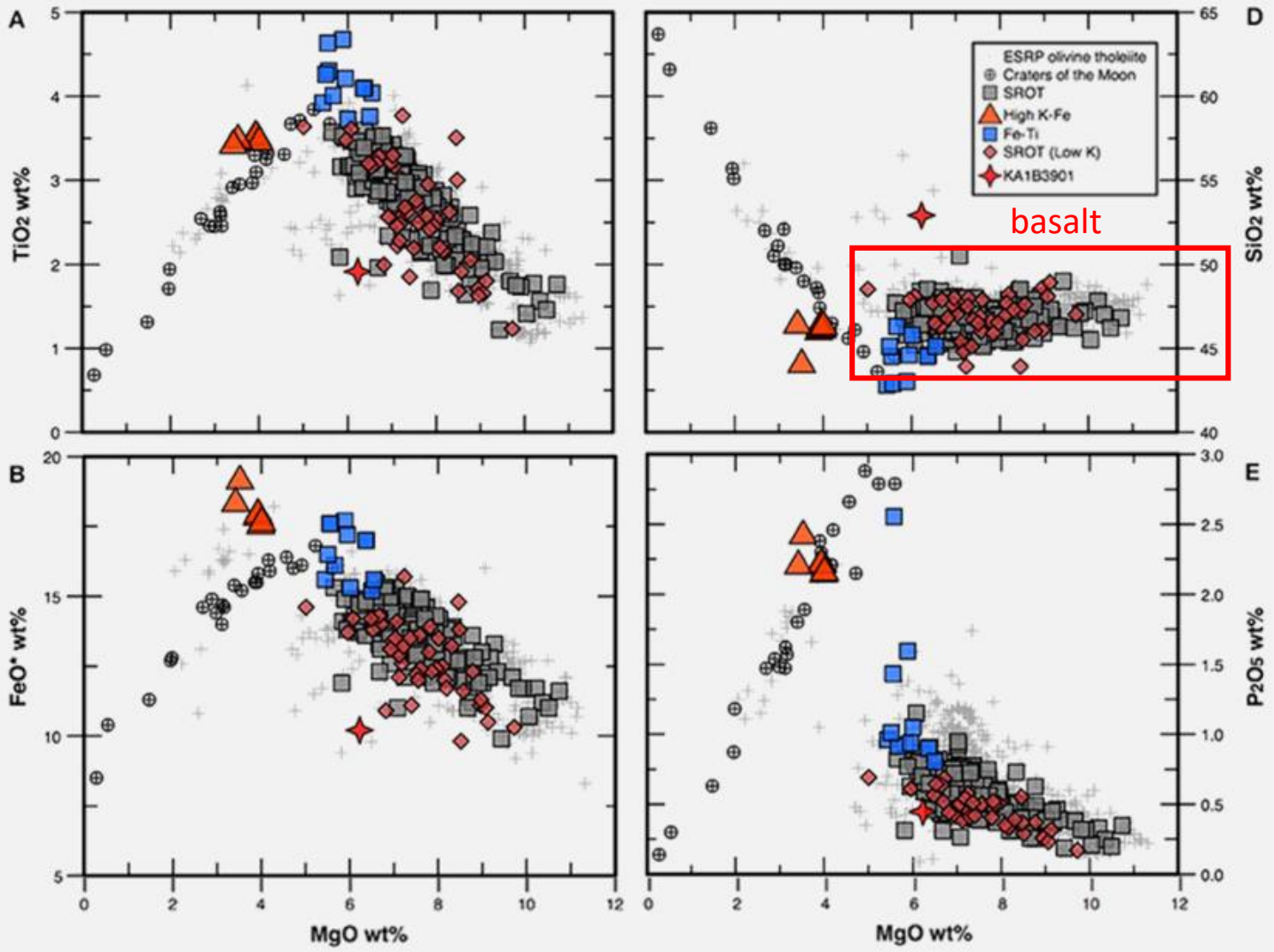
The best choice of x axis is an element or oxide that tracks the process of magma compositional evolution through crystallization.

$SiO_2$ on x axis: good for igneous rock suites that range from basalt through to rhyolite.



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# Variation diagrams




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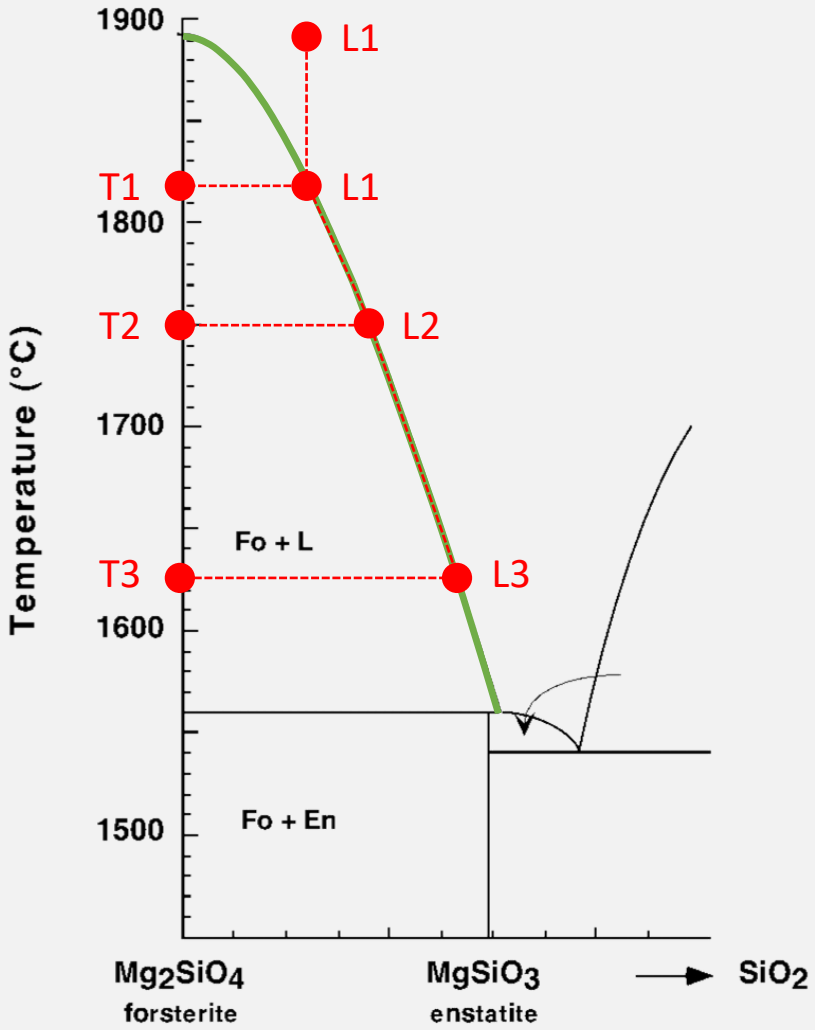
$SiO_2$ on x axis: good for igneous rock suites that range from basalt through to rhyolite.

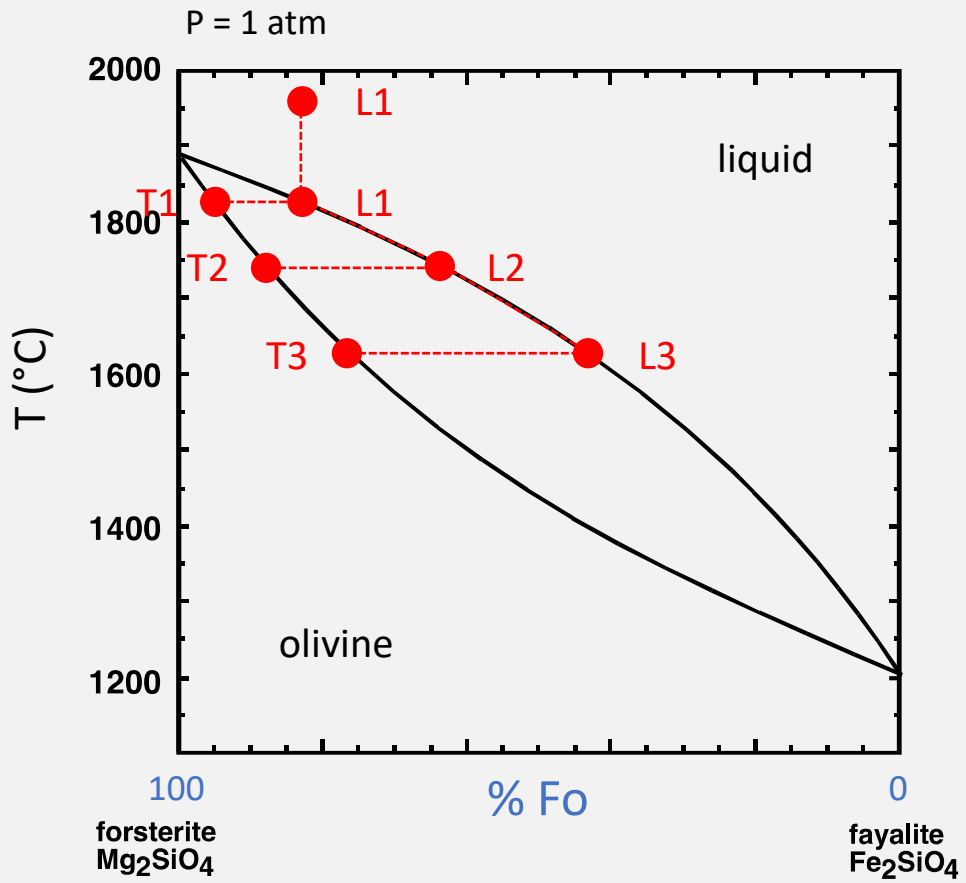
MgO on x axis: good for suites of basaltic rocks where there is little variability in $SiO_2$ content.


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# Liquid lines of descent: Olivine crystallization

See also EART 26201 Principles of Geochemistry, Lecture 3

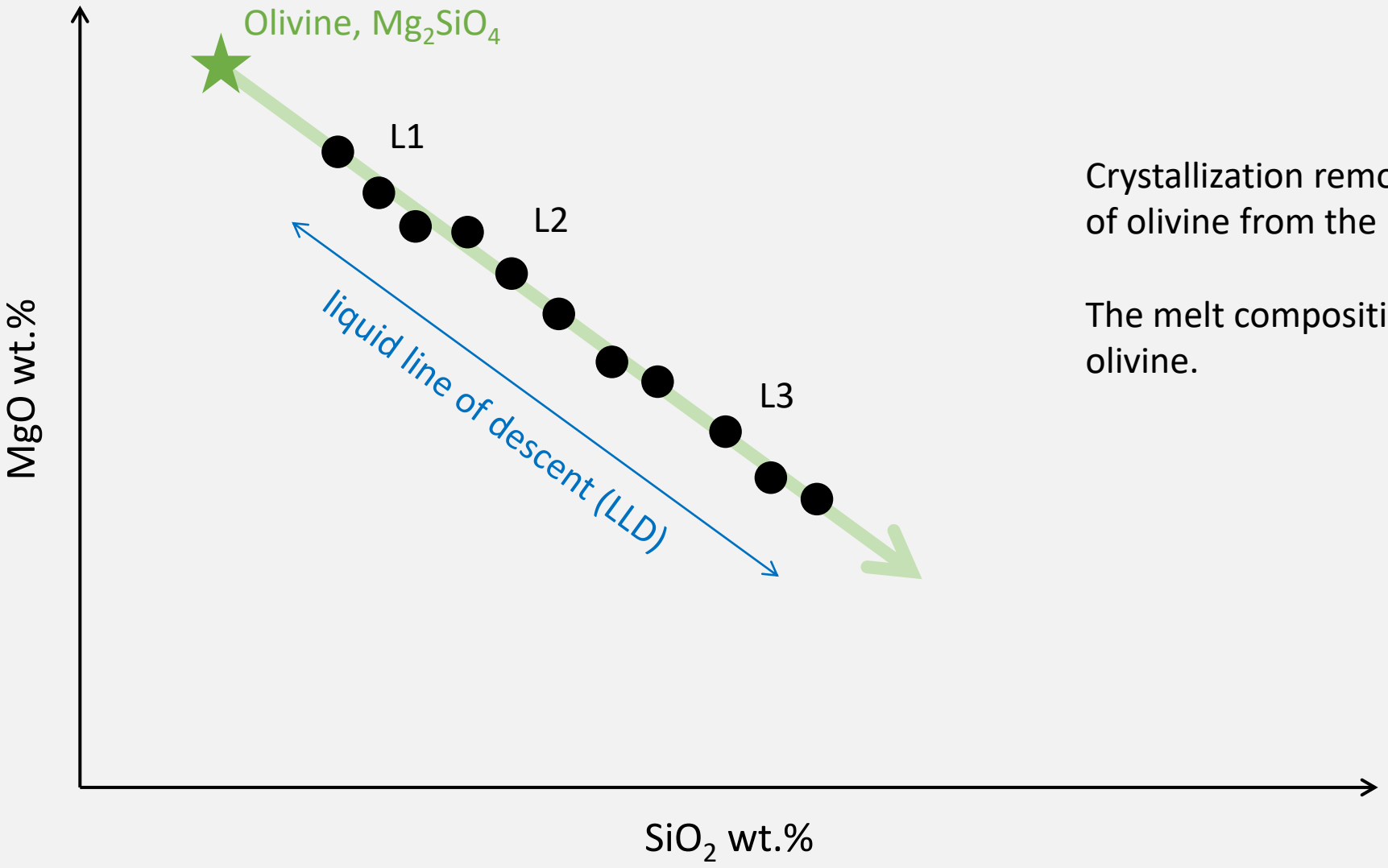





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# Liquid lines of descent: One mineral crystallizing - olivine

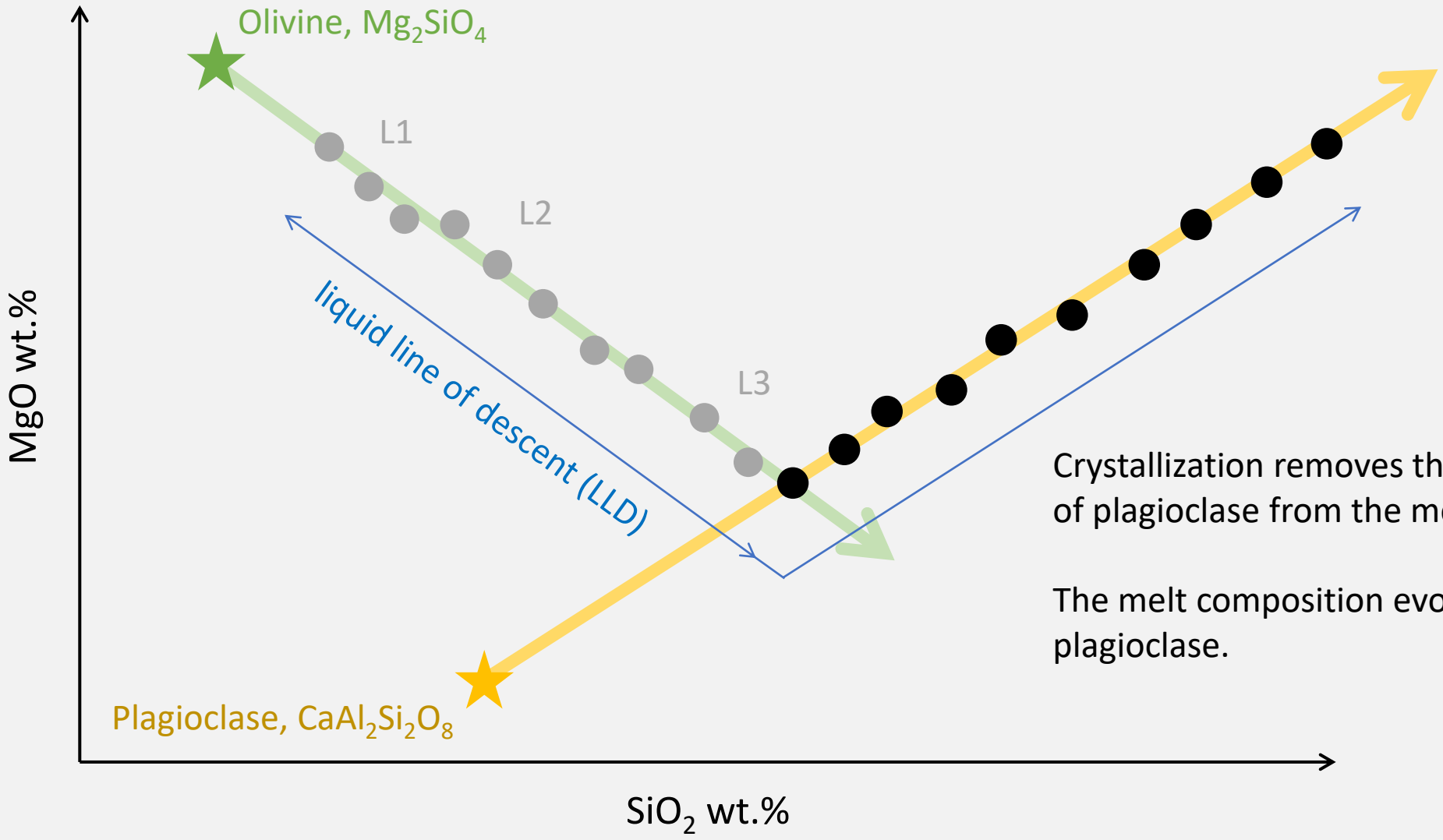



Crystallization removes the chemical constituents of olivine from the melt.

The melt composition evolves directly away from olivine.


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# Liquid lines of descent: One mineral crystallizing - plagioclase

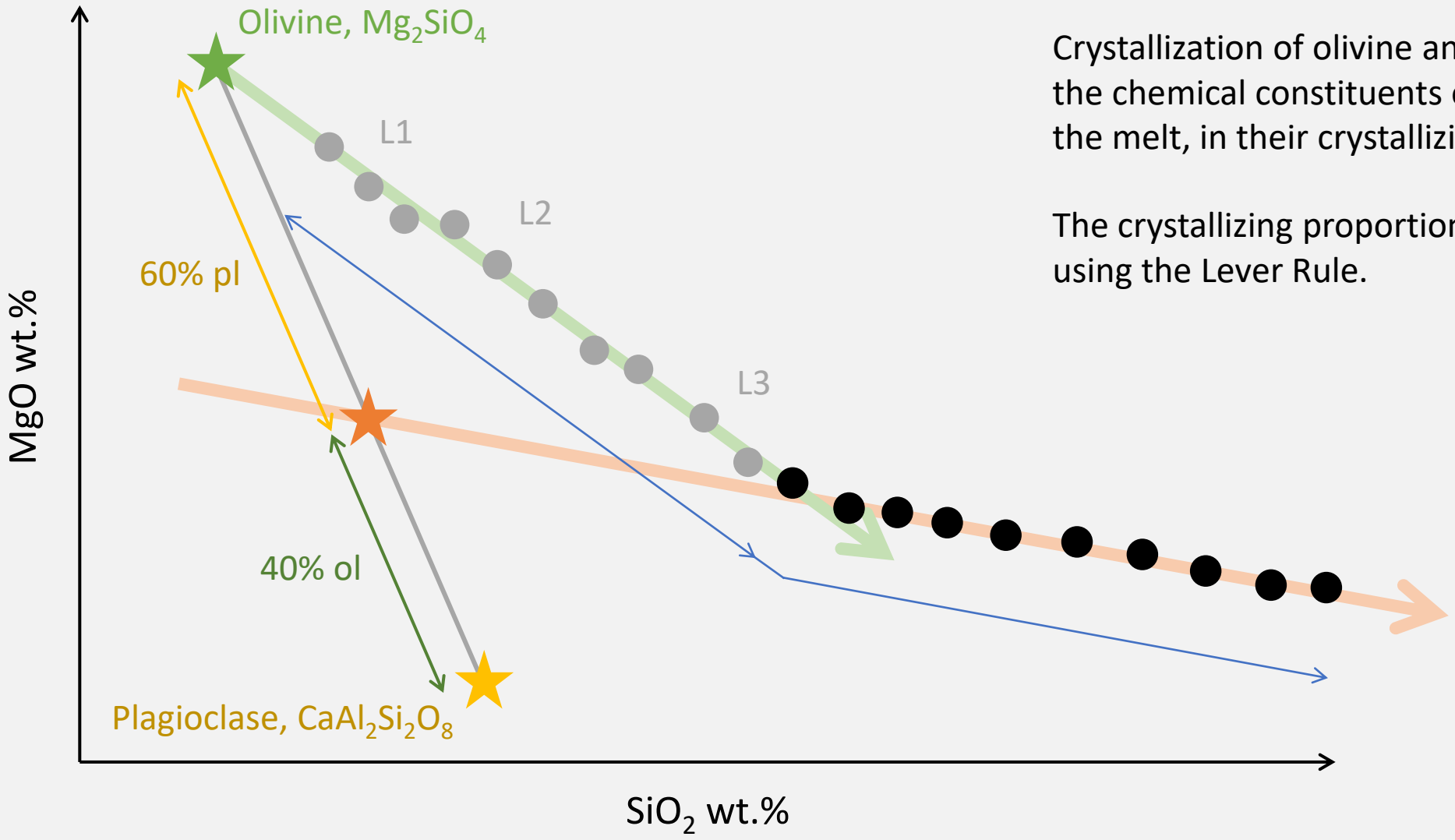



Crystallization removes the chemical constituents of plagioclase from the melt.

The melt composition evolves directly away from plagioclase.


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# Liquid lines of descent: Two minerals crystallizing – olivine and plagioclase

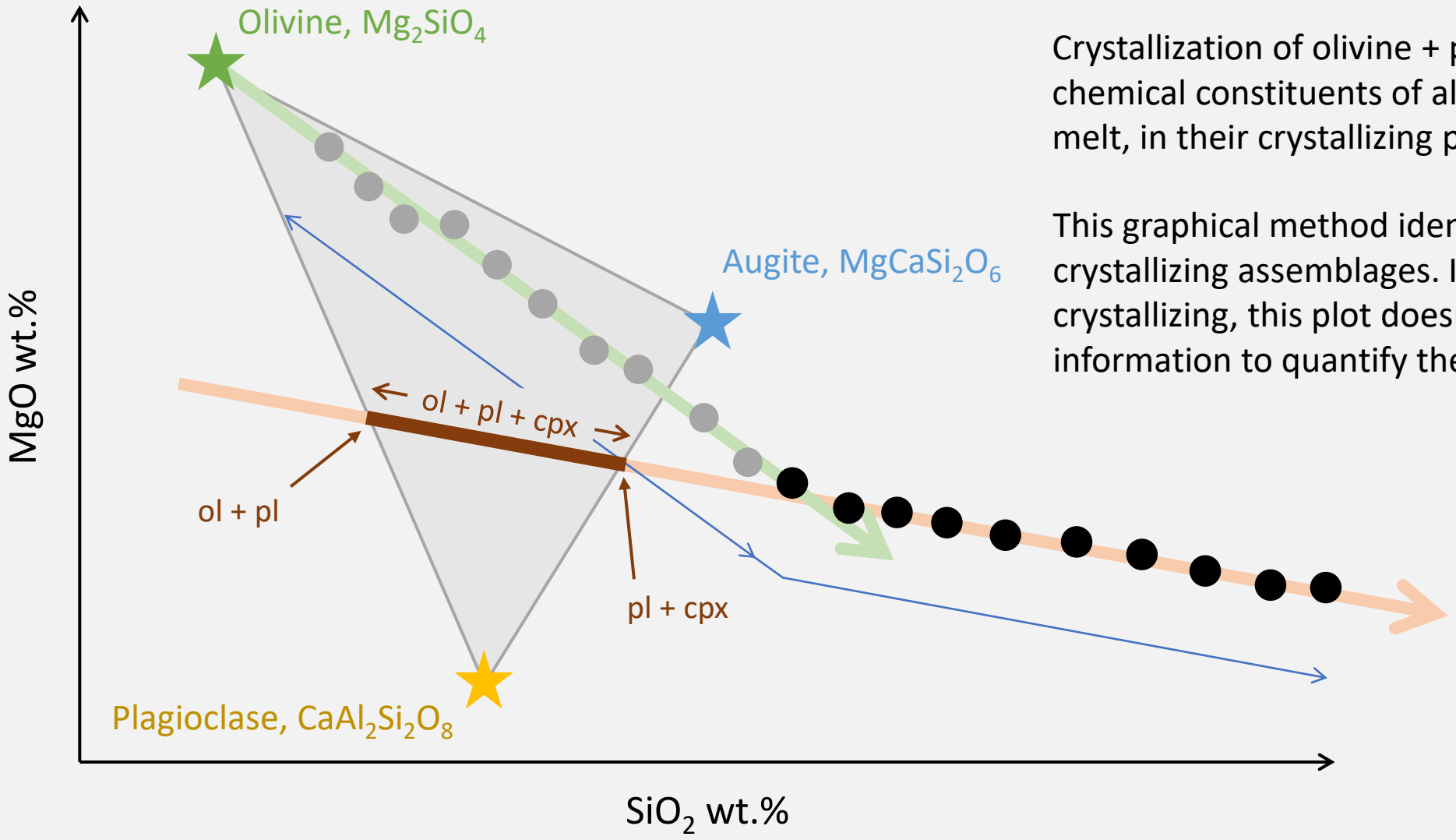



Crystallization of olivine and plagioclase removes the chemical constituents of both minerals from the melt, in their crystallizing proportions.

The crystallizing proportions can be calculated using the Lever Rule.


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# Liquid lines of descent: Three minerals crystallizing – olivine, plagioclase and clinopyroxene




Crystallization of olivine + plag + cpx removes the chemical constituents of all three minerals from the melt, in their crystallizing proportions.

This graphical method identifies three possible crystallizing assemblages. If all three minerals are crystallizing, this plot doesn't provide enough information to quantify the crystallizing proportions.

To determine the correct crystallizing assemblage and/or the crystallizing proportions, we could look at a thin section.


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# Igneous processes: Crystallization

You should now be able to:

- Explain the key chemical differences between mafic, intermediate and evolved igneous rocks
- Interpret variation diagrams to provide qualitative descriptions of the crystallization trends in igneous rock suites
- Use variation diagrams to identify changes in the crystallizing assemblage, and predict what the crystallizing assemblage might be




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